BANK OF THAILAND

MONETARY POLICY REPORT

MARCH 2017

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Bank of Thailand
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Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2017

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2017. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

1. Global Economy

Advanced economies continued to recover. This was particularly the case for the U.S. economy where labor market conditions strengthened with investment poised to increase. The euro area economies gained traction through domestic consumption, while the Japanese economy continued to recover at a gradual pace. The Chinese economy was expected to slow down as a result of the ongoing economic reforms; however, the economic recovery in other Asian economies was expected to continue, driven by exports, especially those of electronic goods which benefited from the technological upcycle and the uptrend of Internet of Things (IoT) products. Overall, **trading partner growth was revised upward.** While there were greater upside risks compared with the previous assessment in the case that U.S. economic stimulus measures would turn out to be more effective than expected, **the balance of risks to the trading partner growth forecast remained skewed downward.** This was due to U.S. trade policies, which involved potentially greater risks and uncertainties than the previous assessment. Meanwhile, risks concerning financial stability in China, political uncertainties in Europe, and problems faced by the European banking sector were judged to be unchanged from the previous assessment.

Most central banks continued their accommodative monetary policy, although policy accommodation was expected to phase down due to improvements in the global economic and inflation outlook. Despite the future direction becoming clearer, there remained uncertainty surrounding the timing and pace of U.S. monetary policy normalization. There was also considerable uncertainty about the U.S. administration's economic policies. These uncertainties were important in determining investors' expectations. Consequently, volatilities in global financial markets were to be expected on the back of increased uncertainties, potentially resulting in volatile capital flows both into and out of emerging markets.

Oil prices were projected to decline in the near term before trending upward later in line with the OPEC's agreement to cut production and the global economic recovery. However, taking into account uncertainty over the OPEC's compliance and the possibility of a swift response of the global oil supply to higher prices, the Monetary Policy Committee revised the projection of the Dubai oil price down to 52.3 dollars per barrel in 2017. The oil price was projected to rise to 54.8 dollars per barrel in 2018.

2. Recent Economic Developments

The Thai economy in the fourth quarter of 2016 continued to recover on the back of private consumption and public spending as well as the rebound in merchandise exports. Meanwhile, **headline inflation over the first two months of 2017 continued to rise relative to the rate seen at the end of last year**, due largely to higher energy prices. Meanwhile, core inflation slowed down slightly.

Financial conditions remained accommodative. Long-term government bond yields rose due to external factors while short- and medium-term government bond yields increased only slightly. Business loans showed signs of higher growth, especially working capital loans extended to exporters and agriculture-related commercial businesses. Household loans slowed down in most categories. This was particularly the case for growth in credit card loans that significantly declined after having accelerated due to government stimulus measures during the end of last year. Meanwhile, the baht against the U.S. dollar appreciated in the same direction as regional currencies due to the weakening of the U.S. dollar, as market participants viewed details of U.S. economic stimulus policies still unclear. **Financial stability remained sound overall.** The defaults of bills of exchange (B/E) issued by unrated companies since the end of last year had a limited impact on the financial system and did not affect overall corporate bond issuance.

3. Economic Outlook

In the period ahead Thailand's economic recovery was expected to gain further traction from 2016. The economy was projected to record growth of 3.4 percent and 3.6 percent in 2017 and 2018, respectively.

Export growth showed a steady recovery as seen in many products. The value of merchandise exports was projected to expand by 2.2 percent in 2017 led by (1) electronics products due to the increased popularity of smartphones and IoT devices, (2) electrical appliances that benefitted from improved external demand, and (3) the relocation of manufacturing sites to Thailand of products such as vehicle tires and solar cells. Moreover, export prices of commodities and petroleum-related products were forecast to trend upward along with crude oil prices.

Exports of services experienced a quick recovery and were projected to continue expanding after the impact of measures to curb illegal tour operators subsided. In addition, the extension of visa fee reduction/exemption as well as the Chinese government's restriction on sales of tour packages to South Korea were likely to bring in more Chinese tourists to Thailand. Therefore, the Committee revised up the projection of the number of foreign tourists to 34.5 million in 2017.

Public spending remained an important driver of growth and was expected to record a higher figure than the previous assessment despite a delay in parts of the dual-track railway project due to the revision in the government's procurement processes. This upgraded public spending projection was attributable to clearer investment plans under the Pracharat project, the higher-than-expected fiscal budget for fiscal year 2018, and the faster-than-expected budget disbursement for telecommunications infrastructure upgrade.

Private consumption was expected to continue recovering. Key supporting factors are (1) improvements in farm income due to higher rubber prices and higher rice output, (2) a positive income outlook for households in the service sector, and (3) some increase in purchasing power following the gradual completion of loan repayments under the first-car scheme. However, overall income and employment might have not yet benefitted clearly from the recovery in merchandise exports, as the exports improvements were largely concentrated in the sectors which relied more on capital rather than labor for production.

Private investment showed signs of recovery in some sectors such as telecommunications, electronics, and alternative energy. Although investment might not gain much from the recovery in private consumption and merchandise exports at this stage, as there remained excess production capacity, private investment was expected, going forward, to pick up once consumption and exports growth become firmer.

Headline inflation was projected to gradually rise and approach the midpoint of the target due to the phasing out of the base effect from last year's lower oil prices as well as increased demand pressure from the economic recovery. Nevertheless, headline inflation might fluctuate in the near term due to the base effects of higher fresh food prices during the drought last year. At the same time, downside risks increased, attributable to global oil prices that faced downward pressures due to uncertainties pertaining to global oil supply. **Headline inflation was projected to average at 1.2 and 1.9 percent in 2017 and 2018, respectively.**

The balance of risks to growth and inflation forecasts tilted towards the downside and to a greater extent compared with the previous assessment. This was on account of (1) uncertainty surrounding the U.S. foreign trade policies that could undermine exports and (2) financial stability concerns in China that could slow down China's economic growth further. Nonetheless, there remained a possibility that the Thai economy would grow at a faster rate than the baseline projection if public investment were to crowd in further private investment and if U.S. economic stimulus policies turned out to be more effective than expected.

4. Monetary Policy Deliberation

In the Monetary Policy Committee meetings on 8 February and 29 March 2017, the Committee unanimously voted to keep the policy rate on hold at 1.50 percent. In deliberating their policy decision, the Committee considered that the growth outlook improved and that the economy would expand at a faster pace than the previous assessment. This was on account of a steady recovery in exports and a quick turnaround in tourism. However, the positive spillovers from improvements in exports to employment and income remained limited. Private consumption and private investment, however, continued to recover at a gradual pace. Meanwhile, headline inflation had reached the lower bound of the target and was expected to gradually rise despite a downward revision to the forecasts due to lower inflation outturns and the recent decline in oil prices.

Overall financial conditions remained accommodative. Total corporate financing increased mainly for those businesses with robust recoveries. Meanwhile, the increase in long-term bond yields had a limited impact on financing costs because Thai businesses mainly issued short- and medium-term debt. The baht appreciated somewhat against major trading partners' currencies over the recent period, partly due to Thailand's relatively strong external position. In the period ahead, capital flows and exchange rates would continue to experience heightened volatility, and thus proper foreign exchange risk management on the part of the private sector would become increasingly important.

Financial stability remained sound. However, there remained pockets of risks that warranted close monitoring such as the deterioration in debt serviceability of SMEs and households, yield-seeking behavior that might lead to underpricing of risks, and shadow banking activities together with other risks that potentially remained unnoticed. Moreover, the prolonged period of low interest rates amid the rise in long-term bond yields might induce some businesses to rely more on issuance of short-term bonds which could pose a risk if there were maturity mismatches between assets and liabilities. In addition, collaboration among relevant regulatory agencies would play a greater role in reducing vulnerability buildups in Thailand's financial system.

Monetary policy should remain accommodative until the economy achieved a broader and firmer growth which would facilitate the return of inflation to the midpoint of the target, especially given that there remained uncertainties on the external front particularly those pertaining to the fragile global economic recovery and the U.S. foreign trade policies that might undermine export recovery. The Committee would stand ready to utilize an appropriate mix of available policy tools in order to ensure that monetary conditions remained conducive to the continuation of economic growth while ensuring financial stability.

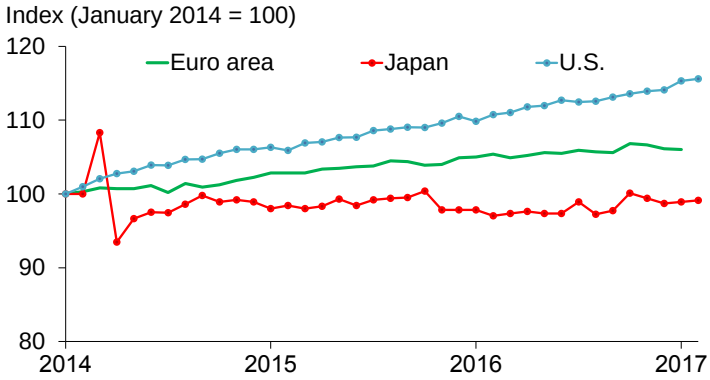
1. The Global Economy

Advanced economies continued to recover, providing support for Thailand's economic growth.

The **U.S.** economic recovery was gaining traction thanks to a significant recovery in the labor market that supported domestic consumption. Moreover, improving business confidence after the presidential election, higher corporate profits, and rising oil prices from their lowest levels in the first quarter of 2016 led to a stronger recovery in private investment. The **euro area** continued to expand, supported by consumption, accommodative monetary policy, and more broad-based recovery in manufacturing. The **Japanese economy** gradually recovered, underpinned by accommodative monetary policy and the gradual recovery in domestic demand.

Chart 1.1 Domestic consumption in major advanced economies expanded, particularly in the U.S.

(retail sale)



Source: Bloomberg

Looking ahead, the recovery of the advanced economies is still facing risks from the uncertainties of the U.S. policies on international trade, tax policy reform, and infrastructure investment. For the euro area, issues warrant close monitoring include political uncertainty especially the elections in France and Germany this year, problems in the European banking sector, and the Brexit process.

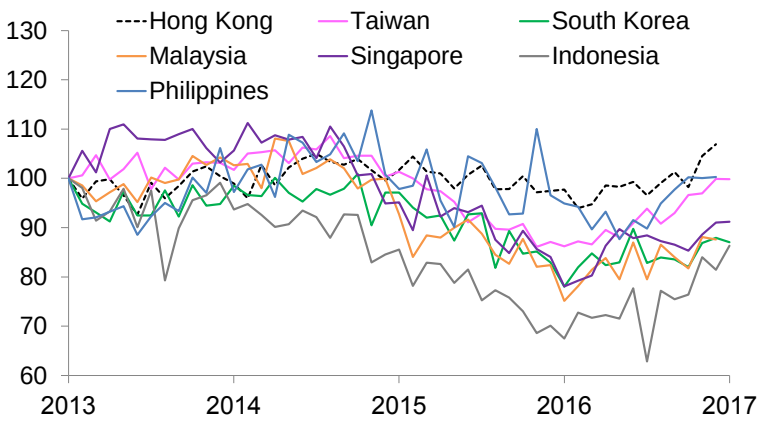
China's growth slowed down following structural reform, while the Asian economies recovered as exports, especially of electronics, improved.

The **Chinese economy** slowed down following the government's economic structural reform. The reform plan places more emphasis on long-term economic stability by turning to economic development on quality rather than quantity, reduction of excess capacity on production, and less dependence on government investment. The Committee expected the Chinese economy to grow this year in line with the government's 6.5 percent target and expand by 6.2 percent in 2018. The Committee saw a need to closely monitor economic and financial stability risks in China, particularly the high level of corporate debt, speculative activities in the real estate sector, and capital outflows.

Asian economies (excluding Japan and China) recovered gradually, supported by the continued export recovery in the fourth quarter of 2016 and better-than-expected latest economic indicators in the first quarter of 2017. In particular, export of electronics products benefitted from the upturn of the technology cycle (Box: *Implications of the electronics technology life-cycle and the rise of the Internet of Things for Asia's exports)* while exports of petroleum-related products also gained from higher oil prices. The export recovery already had some positive effects on investment and employment in countries that are highly dependent on exports. Meanwhile, domestic consumption increased gradually. Nevertheless, the Asian economic recovery in the period ahead still faces risks from (1) uncertainty in the U.S. trade policy which may impact on trade and investment climate in the region; (2) elevated household debt and tighter liquidity weighing on consumption in some countries; and (3) changes in the global trade structure that can restrain the Asian export recovery. In particular, developed economies turn more to domestic production instead of foreign-based production, or relocate production bases from Asia to lower-cost areas closer to their markets such as Eastern Europe and Mexico.

Chart 1.2 Recovery of Asian exports

Seasonally adjusted index of export value (January 2013 = 100)



Source: CEIC

Growth prospects for Thailand's trading partners improved, but downside risks remained.

Economic growth of most Thailand's trading partners was better than expected in the fourth quarter of 2016 and would likely gain traction in the period ahead. The Committee revised the growth assumption for trading partners to 3.3 percent for 2017 and 2018. Major advanced economies, especially the U.S., showed stronger growth, while Asian exports recovered more firmly. Upside risks slightly increased as the U.S. economic stimulus policy could be implemented more than expected. Moreover, the investment cycle improved, particularly in the U.S., and the spillovers of improvements of Asian exports on domestic economic activities could turn out to be greater than expected. However, downside risks still weighed on the global economy, as the impact of the U.S. trade policy could affect China's and Asia's highly interlinked exports. Risks regarding financial instability in China and political uncertainty as well as problems in the banking sector in Europe remained.

Table 1.1 Assumption on trading partners' economic growth

Annual change (%YoY)	Weight (%)	2016*	2017	2018
United States	14.9	1.6	2.3 (2.2)	2.4
Euro area	10.0	1.7	1.5 (1.1)	1.5
Japan	13.6	1.0	1.1 (1.1)	0.9
China	15.7	6.7	6.5 (6.3)	6.2
Asia (excluding Japan and China)**	37.4	3.5	3.6 (3.5)	3.8
Total***	100	3.1	3.3 (3.1)	3.3

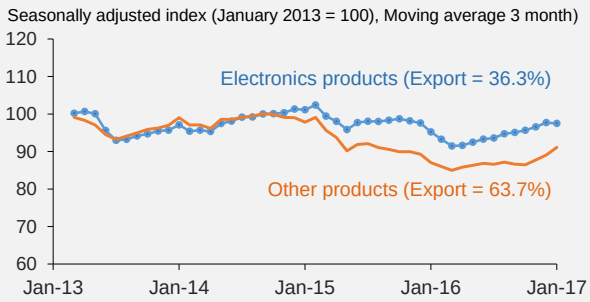
Note: *Outturn
**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by a share of Thailand's total exports to major 13 trading partners as of 2014 (including the United Kingdom and Australia)
( ) reported in Monetary Policy Report December 2016

Implications of the electronics technology life-cycle and the rise of the Internet of Things for Asia's exports

Asia's exports continued to recover in the second half of 2016 following a prolonged period of sluggish growth. Commodities, food, and electronics products were key growth drivers, supported by three major factors: rising commodity prices, improving household purchasing power and consumer confidence in several economic regions, and the recovery cycle of electronics products. Going forward, Asia's exports will continue to expand thanks to the turn of the electronics products life cycle. Key supporting factors for the upturn in electronics exports are:

Chart 1 Asia's exports*are recovering especially for electronics products

Exports by product



Note: * Asia including China, Hong Kong, Taiwan, South Korea, Malaysia, Singapore and Thailand (excluding Indonesia and the Philippines due to delayed data release)
Sources: CEIC by the Bank of Thailand's calculation

(1) Intense competition in the electronics market has led to advancement of technology. Examples include the use of fingerprint scanners on electrical appliances other than smart phones, small USB type-C that can transmit information at a faster rate, curved organic light emitting diodes (OLED) screen in smartphones and TVs, dual cameras and improved LTE signal receptors in smartphones. This trend induces demand for these electronics parts.

(2) Smartphones markets will continue to expand. Markets in developing countries such as India, Indonesia, and the Philippines still have a large room to grow because of their currently low penetration rates. Furthermore, consumers' desire to keep up with new technology will drive more demand for smartphones. A research company, Statista, estimates that worldwide smartphone users will increase by 10% and 9% in 2017 and 2018, respectively. Going forward, this trend will support growth for smartphones and parts.

(3) Growth of the Internet of Things (IoT) will accelerate. The IoTs refer to electronics aside from smartphones that can connect to other appliances. Examples include wearables, medical devices, household appliances, as well as other equipment used to improve productivity in the manufacturing and agricultural sectors. These devices have capability to record user preferences and process information to enhance users' daily-life convenience. Another research firm, Gartner, estimates that the IoT technology is still in its early development stage and possesses a potential to grow at a fast pace of 30-35% annually. By 2020, the production level of IoT devices could reach over 20 billion from the current level of 6-7 billion devices per year. This consequently translates to rising demand for electronics parts as well.

The above factors suggest that exports of electronics from Asia including Thailand will steadily grow. As for Thailand, the public and private sectors must collaborate on development of the electronics industry, especially on research and development of new outputs that responds to global demand such as IoT products. Doing so will provide a boost to Thai exports to fully benefit from the rapidly rising global demand for electronics in the future.

Most central banks maintained accommodative monetary policy but with a lower degree of easing going forward.

Most central banks maintained accommodative monetary policy but with a lower degree of easing going forward, on the back of improvements in the global economy and higher inflation. Over the recent period, demand-pull inflationary pressures increased in some countries, especially in the U.S. where it approached full employment amid continuing stimulus fiscal policy. This led the Fed to raise the policy interest rate at the recent meeting on March 14-15, 2017 by 25 basis points to 0.75-1.00 percent. This rate hike was earlier than previously expected at the beginning of the year. Further increases were anticipated in the period ahead. Other central banks could also gradually reduce monetary policy accommodation as inflation increased.

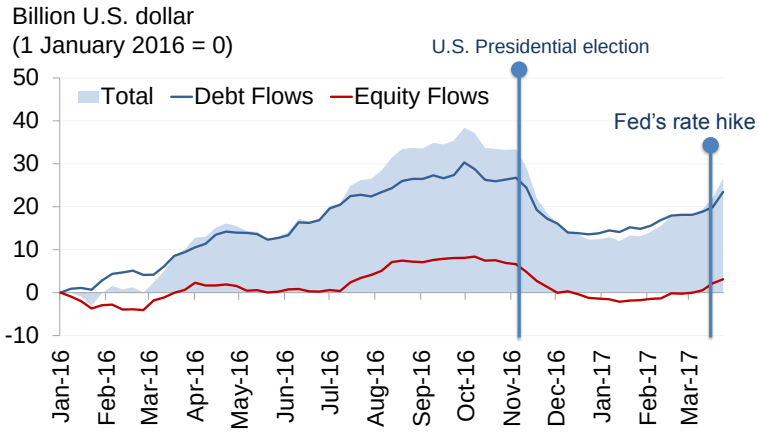
Volatility in the global financial markets rose as a result of increased uncertainty stemming from the U.S. economic policy, political concerns in Europe, and financial instability in China.

In late 2016, the outcome of the U.S. presidential election and the Fed's policy rate hike induced a net capital outflow from emerging market economies (EMEs) back to the U.S. Later in the beginning of 2017, however, uncertainty in the U.S. economic stimulus policy led to adjustments in investor expectations and a return of investment in EMEs. Furthermore, investor expectations on key global factors, such as the pace of Fed rate hikes, could lead to changes in expectations on asset prices from time to time. In the first half of March, capital inflows to EMEs slowed down, as investors expected the U.S. policy interest rate to increase earlier than previously anticipated at the beginning of the year. On the other hand, for the latter half of March after the Fed's rate hike, capital inflows to EMEs accelerated as investors revised their expectations on the U.S. policy interest rate increase to be more gradual than previously expected. In addition, the U.S. administration was unable to pass the new American Health Care Act, adding more uncertainty regarding the ability to push for other policy initiatives.

Looking forward, global financial markets remain highly volatile with fluctuations in cross-border capital movements due to uncertainty in U.S. economic policy, European political instability, and banking sector problems in both Europe and China.

Chart 1.3 Foreign investors returned to the emerging markets since the beginning of 2017. Foreign investment slowed down in the beginning of March before speeding up after the Fed's rate hike as investors' expectations were adjusted.

Accumulated capital inflow to emerging markets*



Note: *Thailand, Indonesia, India, South Africa and Turkey (Turkish data as of 15 March 2017)

Source: Institute of International Finance

Oil prices increased during the first two months of 2017 but could decline temporarily before gradually trending up in line with the global economic recovery.

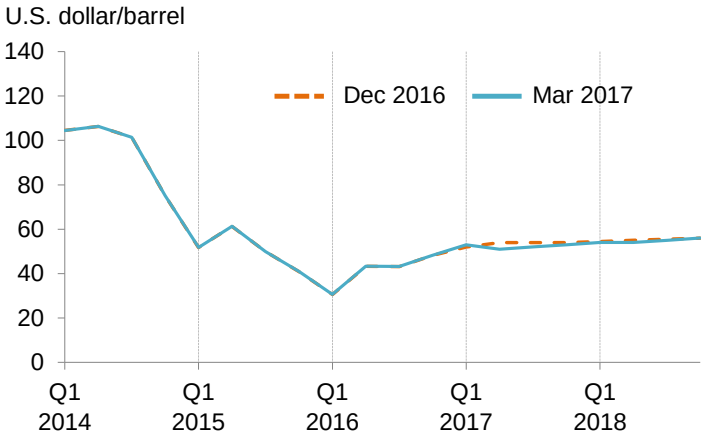
Prices of Dubai crude oil increased, as the OPEC cut production, but started to fall recently given the uncertain U.S. supply. In November 2016, the OPEC reached an agreement to cut production and also managed a compliance rate of 93 percent, higher than their historical level and market expectations. Rising demand from developing counties also pushed crude prices up to more than 50 U.S. dollars per barrel during the beginning of 2017. However, oil prices started to come down in March because the increase in supply was more than expected, especially those from the U.S.

For the period ahead, crude prices are expected to decline temporarily before gradually trending up in line with the global economic recovery and higher demand. Stock accumulation remained at a high level, especially in the U.S. Shale oil producers in the U.S. could quickly return to pump more supply. These factors continued to pressure oil prices and weigh on their recovery in the future. As a result, the Committee revised the assumption on Dubai oil prices down to 52.3 and 54.8 U.S. dollars per barrel in 2016 and 2017, respectively. Risks to the assumption were adjusted to skew downward instead of a balanced risk in the previous assessment. This was mainly due to uncertainty in future compliance among OPEC producers to the agreed production level as well as oil production in the U.S. Meanwhile, unrests in the Middle East and Africa such as the nuclear tests in Iran and terrorism in Nigeria and Libya carry risks that might bring oil prices higher than anticipated.

Metal prices were expected to surge over the short period due to greater demand from improving market confidence given the U.S. economic policy, oil price recovery, and infrastructure investment in China. On the supply side, production was lowered in the short term due to problems in major producing countries, namely disputes between copper companies and the Indonesian government, together with production cut in four large Chinese provinces. Looking ahead, metal prices would only slightly increase after having accelerated this year.

Chart 1.4 Oil prices decline temporarily before gradually trending up along with the global economic recovery.

Assumption on Dubai oil price



2. The Thai Economy

2.1 Recent developments

The Thai economy continued to recover on the back of private consumption, public spending, and the rebound in exports of goods.

The Thai economy in the fourth quarter of 2016 expanded 3 percent from the same period last year. The key growth drivers were private consumption, supported by higher farm income and government stimulus measures, expedited disbursement of the fiscal budget, and the resumption of growth in exports of goods following prior contractions. These drivers combined to support growth momentum and helped offset declines in private investment and the slowdown in exports of services that resulted from measures implemented to curb illegal tour operators. Overall, the Thai economy grew 0.6 percent from the previous quarter after seasonal adjustments. For the whole of 2016, the Thai economy expanded 3.2 percent, an improvement from 2.9 percent in 2015.

Thailand's economic growth continued to gain traction over the first two months of 2017. Private consumption recovered at a gradual pace and was partly supported by the launch of new car models, while public spending continued to provide growth momentum. Private investment remained flat, however. In any case, rising external demand appeared to contribute to the swift recovery in tourism that was previously affected by measures to curb illegal tour operators. Stronger external demand also led to the increasingly broad-based growth in export of goods such as certain agricultural products, electronics, electrical appliances, automotive and parts, as well as petroleum-related products, all of which accounted for about 40 percent of total exports. Additionally, the improvement seen in the exports of electronics from Thailand was also supported by the technological upcycle and uptrend of Internet of Things (IoT) products and was in line with exports of electronics from the region. Further growth opportunities was expected in the period ahead.

The recovery in exports of goods had limited positive effects on employment, while household income increased slightly for some household groups.

The recovery in exports of goods had limited positive effects on employment because (1) the recovery was still at the early stages, (2) most of the improvements seen were in capital-intensive industries, such as the electronics industry which accounts for only 1.2 percent of total employment, and (3) increasing adoption of automation in the manufacturing processes by some industries in response to labor shortage and to enhance productivity.

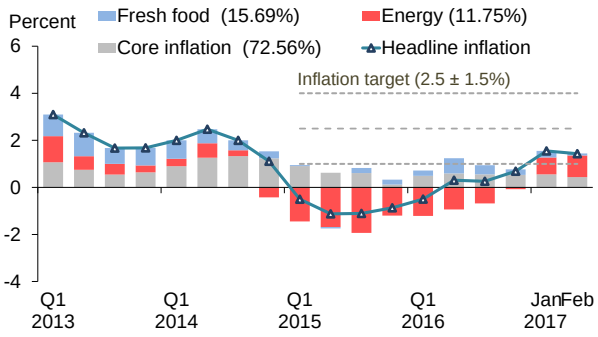
Household income increased slightly but the increase was not broad-based and thus did not spur private consumption as much as it should. For instance, the increase in farm income due to higher rubber prices was limited to rubber farmers in the eastern and the northeastern regions, while rubber farmers in the south suffered from lower rubber output due to the floods. Meanwhile, rice prices were stable but remained at low levels. Meanwhile, the recovery in non-farm income remained subdued overall despite increases due to higher employment in some service sectors, namely wholesale and retail trade, because wages in these sectors were rather low.

Headline inflation[1/] rose from the end of last year and returned to the lower bound of the inflation target on account of higher energy prices, while core inflation declined slightly.

Headline inflation in the first two months of 2017 increased from the end of last year, averaging at 1.49 percent. The increase was attributable to (1) the increase in retail oil prices due to higher global crude prices and (2) the base effect of last year's lower crude prices. Meanwhile, fresh food prices fell on account of lower meat prices as well as higher output of vegetable and fruits. The one-year-ahead inflation expectations of businesses were roughly around the previous figure at 2.1 percent, while medium-term inflation expectations of the public remained close to the midpoint of the target at 2.5 percent.

Chart 2.1 Headline inflation rose from the end of last year and returned to the target

Headline inflation and inflation target

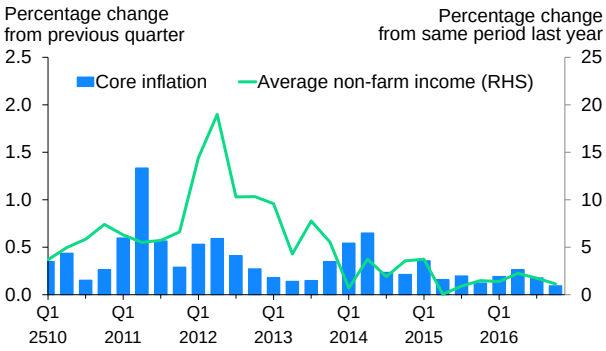


Note: () denotes share in inflation basket

Source: Ministry of Commerce, calculation by Bank of Thailand

Chart 2.2 Average non-farm income saw subdued recovery as demand-pull inflationary reassures remained low.

Core inflation and growth of non-farm income



Source: National Statistics Office and Ministry of Commerce

Core inflation in the first two months of 2017 slowed down from the previous quarter, averaging at 0.67 percent. The main reason behind the slowdown was the base effect of last year's higher tobacco prices due to an increase in excise tax on tobacco, as well as subdued demand-pull inflationary pressures given the gradual pace of the domestic economy recovery and low farm income. At the same time, cost-push inflationary pressures rose somewhat from higher oil prices but the results of the Business Sentiment Survey[2/] revealed a higher ratio of respondents that indicated greater difficulty in raising prices, suggesting the existence of short-term constraints faced by producers in making corresponding price adjustments.

Table 2.1 Inflation

Percent	2015		2016				2017
	Q3	Q4	Q1	Q2	Q3	Q4	Jan - Feb
Headline CPI	-1.11	-0.86	-0.50	0.30	0.26	0.69	1.49
Core CPI	0.92	0.85	0.67	0.79	0.76	0.73	0.67
Fresh food	1.48	1.45	1.52	4.24	2.58	1.54	1.20
Energy	-16.42	-14.63	-11.41	-8.95	-7.00	-1.06	8.07

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

[1/] The Ministry of Commerce changed the base year for CPI from 2011 to 2015

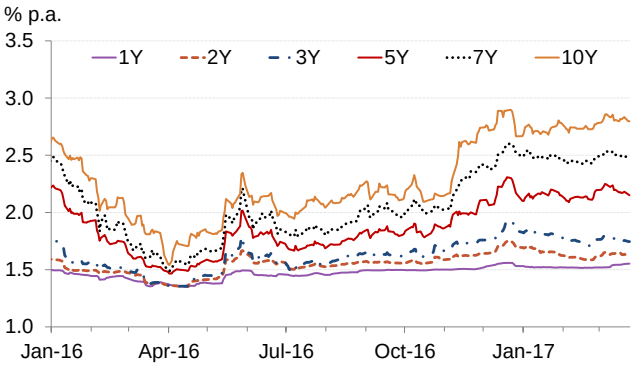
[2/] The Business Sentiment Survey covers about 600 business operators per month across the country

Government bond yields increased due to external factors.

Short-term money market rates in the first quarter of 2017 remained close to the policy rate. **Government bond yields** in the beginning of the first quarter declined from the end of 2016 mainly as a result of concerns over political uncertainties in Europe and unclear direction of U.S. economic policies. Since early March, government bond yields edged up following interviews of several members of the Federal Open Market Committee (FOMC) which prompted market expectations that the FOMC would decide to raise the fed funds rate in its meeting in March. After the FOMC did raise the fed funds rate in its March meeting, however, Thai government bond yields fell as the accompanying statement to the decision was more dovish than the market expected. On the whole, short- and medium-term bond yields remained below levels at the end of 2016, while long-term yields increased.

Chart 2.3 Government bond yields rose in response to external factors

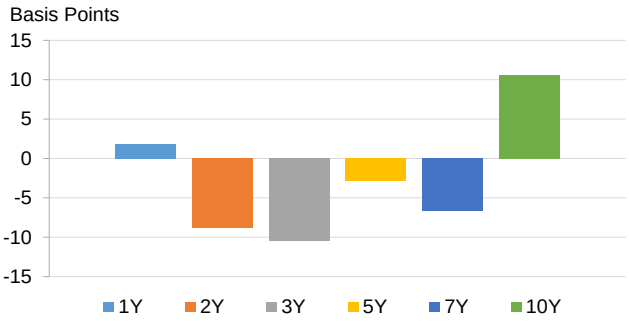
Government bond yields



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 2.4 Short-term and medium-term bond yields were lower than at the end of 2015, but long-term yields increased.

Changes in government bond yields relative to the end of 2016
(As of March 28, 2017)



Source: Thai Bond Market Association (Thai BMA)

Corporate bond yields trended up alongside increases in long-term government bond yields and credit spreads, which tightened financial conditions somewhat. However, the impact on the private sector's financing costs appeared limited as debt issued by most Thai businesses were of 1-3 years maturity, the yields of which increased only slightly and remained below historical averages.

Growth in private credit improved with signs of increase in some business sectors.

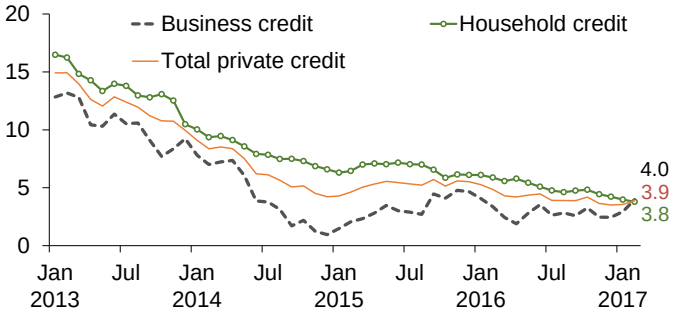
Business credit, especially those extended to export-related sectors, started to show signs of growth, while household credit continued to expand albeit at a slower pace. Consequently, private credit overall at the end of February 2017 grew 3.9 percent from the same period of last year, higher than the 3.5 percent growth recorded at the end of the fourth quarter of 2016.[3/]

[3/] Outstanding credit of other depositary corporations (ODC) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds.

Chart 2.5 Growth in private credit

Growth of private credit

Percentage change from same period last year



Note: Outstanding credit of other depository corporations (ODC) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

Source: Bank of Thailand

Change in business credit outstanding began to show signs of faster growth, although growth remained concentrated in export-oriented businesses, particularly those related to exports of rubber and plastic, which was in line with improvements seen in the export sector. Working capital credit extended to exporters in the first two months of 2017 on average grew 8.5 percent from the same period last year which was an acceleration from just 0.8 percent growth recorded in the fourth quarter of 2016. On the other hand, growth in working capital credit extended to domestic businesses and investment credit were more subdued. Business credit extended to the commercial sector saw improved growth, especially among businesses related to agricultural raw materials. As for other financing channels, financing through short-term debt instruments softened somewhat in January, especially those issued by unrated companies as reflected by the decline in the rollover ratio of unrated bonds. Financing through the equity market also slowed down slightly in February where most new equity were issued by factory and warehouse development/rental businesses.

Change in household credit outstanding continued to decelerate as private consumption growth was not particularly strong. Growth in consumer credit extended by commercial banks in the first two months of 2017 decelerated, especially credit card loans which previously accelerated due to government stimulus measures rolled out in December 2016. Car loans, however, increased slightly following launches of new car models.

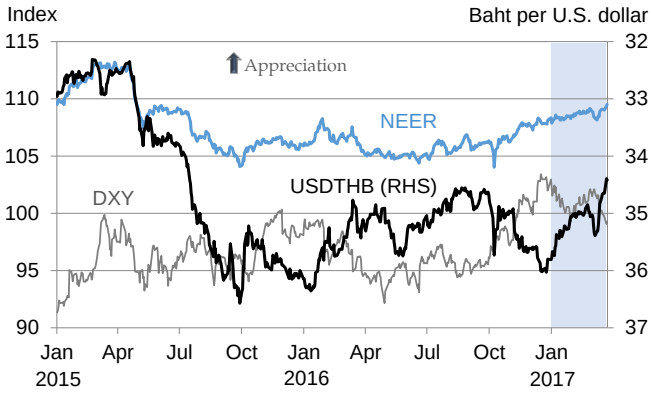
Going forward, financial conditions were expected to remain accommodative but could tighten somewhat as the Fed continues to raise the fed funds rate. In response to such developments, Thai bond yields might increase, although they would likely remain below their 2015 averages. In addition, credit demand was still soft, as reflected in the credit utilization rate which remained close to the level seen in the previous quarter. According to the survey of credit conditions and outlook[4/], financial institutions would still maintain caution in extending credit throughout the first quarter of 2017, especially to borrowers whose credit quality declined in recent periods.

[4/] Survey of credit conditions in the fourth quarter of 2016 and outlook for the first quarter of 2017.

The baht appreciated against the U.S. dollar in line with movements of regional currencies.

From the fourth quarter of 2016 to the first quarter of 2017, **the baht strengthened against the U.S. dollar** as with most regional currencies. Such appreciation resulted from a weaker U.S. dollar due to lack of clarity regarding U.S. expansionary economic policies and negative comments made by the U.S. president on the U.S. dollar's strength. Meanwhile, market expectations regarding the Fed's monetary policy normalization path led to greater volatility in the value of the baht as well as other regional currencies. Consequently, during the first half of March 2017, the baht and regional currencies weakened against the U.S. dollar as market participants viewed monetary policy normalization in the U.S. to proceed faster than previously expected. In the latter half of March 2017, however, the baht and regional currencies resumed appreciation against the U.S. dollar as the market reassessed the rise in the fed funds rate to be more gradual, and as the U.S. government failed to pass a new health care bill. It was also observed that these changes in the expectations of foreign investors affected Thailand more than other countries, especially in terms of portfolio inflows. The baht closed at 34.42 on March 28, 4.0 percent stronger than at the end of the previous quarter. Meanwhile, the nominal effective exchange rate (NEER) closed at 109.52, up 1.3 percent from the end of the previous quarter, mainly due to the baht's appreciation against the U.S. dollar and the yuan[5/]. Going forward, international capital flows and exchange rates would likely remain volatile due to prevailing external uncertainties, especially those related to the timing and pace of monetary policy normalization in U.S. and the clarity of the U.S. expansionary economic policies.

Chart 2.6 The baht strengthened against the U.S. dollar as the U.S. dollar weakened across the board



Source: Bank of Thailand and Reuters (latest data on March 28, 2017)

Financial stability remained sound, but there still were pockets of risks that continued to warrant close monitoring such as debt repayment ability of households and SMEs, excess supply of housing in some areas, and the search-for-yield behavior.

Thailand's economic and financial stability remained sound. Financial institutions maintained high levels of capital buffers and provisions for loan losses to cushion against risks stemming from declining credit quality. Financial positions of listed companies overall projected a more positive outlook. Meanwhile, Thailand's external stability remained strong and able to provide cushion against volatilities in global financial markets. Nevertheless, there were still pockets of risks that continued to warrant monitoring. One such risk was the debt repayment ability of households which remained fragile because of the elevated levels of debt, even as debt repayment ability continued to improve thanks to decelerating household debt and increasing farm income. Nevertheless, household debt showed signs of improvements overall as the ratio of household debt to GDP declined from 81.2 percent in the fourth quarter

[5/] The baht appreciated against the yuan because since the end of 2015 the yuan moved in tandem with the U.S. dollar.

of 2015 to 79.9 percent in the fourth quarter of 2016. On the other hand, debt repayment ability of SMEs was declining because the economic recovery were not yet broad-based. Additionally, the property market's ability to absorb condominium stock also warranted monitoring as demand for housing would likely stay flat after the government's real estate stimulus measures expired, while excess supply of housing, especially condominium units in certain areas and price ranges, remained large. This could impact debt repayment ability of real estate developers with weaker financial positions.

The search-for-yield behavior remained present in the low interest rate environment. Notably, investment in foreign investment funds (FIFs) continued to grow; however, most FIFs invest in investment-grade securities of countries such as the United Arab Emirates, Hong Kong, Qatar, Singapore, China, and Luxemburg. Meanwhile, investment in money market mutual funds that appeared similar to saving accounts slowed down since the fourth quarter of 2016. In addition, recent defaults of unrated bills of exchange since the end of last year had limited impact on the financial system, but some companies with fragile financial standings were prone to default on their debt repayment and thus continued to warrant close monitoring. (Box: *Developments in the unrated bonds market and implications for financial stability stability*)

Developments in the unrated bond market and implications for financial stability

Rapid growth of unrated bonds and the series of defaults in unrated bills of exchange (B/E) in the past 3-4 months have drawn the attention of the MPC and warranted close monitoring. Such development was also part of the MPC's assessment of financial stability risks that could arise from the search-for-yield behavior in the low interest rate environment.

Recent developments in the unrated bond market were found not to affect Thailand's financial stability for several reasons. First, **the size of the unrated bond market remained small.** Despite its rapid expansion from a low base, unrated bonds accounted for only 7.4% of total private bonds outstanding[6/]. Furthermore, the total value of recent unrated bond defaults accounted for just 0.08% of total private bond outstanding. Second, **regulations set by the Securities and Exchange Commission (SEC) limited investment in unrated bonds to a small group of accredited investors (AI)**, namely institutional investors and high-net-worth individuals who have the ability to absorb a significant amount of risks. If general retail investors were to comprise a lion's share of unrated bond holders, the impact of the recent defaults could be more severe and widespread. Third, **recent defaults had limited spillovers to other parts of the financial institutions system**. There were only three asset management firms that invested in the defaulted unrated bonds through AI funds, and their combined default value accounted for only 0.035% of the total net asset value of mutual funds. Additionally, the money invested mainly came from investors' own savings rather than loans on margin accounts. Therefore, such a loss did not affect stability of other financial institutions. Finally, **market adjustments following the defaults proceeded in an orderly fashion without any impact on the overall functioning of the private bond market**. Issuers of unrated bonds maturing in January 2017 were still able to roll over their debts, while other corporates were also able to obtain financing through bank loans, capital markets, and other financing channels. Nonetheless, recent defaults provoked greater caution on the part of investors and had led to a repricing of risks as reflected in the shrinking portion of new short-term bond issuance of both rated and unrated bonds. The BOT's preliminary analysis found that market participants were able to price bonds more accurately as reflected by the

[6/] Data as of January 2017. Data included total private sector bonds in Thailand both registered and not registered with the Thai Bond Market Association (TBMA). If consider only private sector bonds registered with the TBMA, unrated bonds accounted for 4.5%.

higher costs of funds for businesses with weaker financial positions relative to those with a stronger position. Such development was welcomed for, as it would reduce risks stemming from businesses' overreliance on short-term funding as well as underpricing of risks on the part of investors.

Despite its limited impact, the recent defaults of unrated B/Es gave an important lesson for all stakeholders to adjust and exercise greater caution to avoid underpricing of risks, which might lead to the accumulation of imbalances that could undermine financial stability in the future. In addition, **bond issuers** should also be mindful of maturity mismatches and rollover risks by avoiding the use of short-term funding for longer-term projects, and maintaining sufficient liquidity cushions as well as access to alternative funding sources in the event that debt rollover were not possible. **Financial intermediaries** that bridge the gap between investors and bond issuers, such as asset management and securities companies, should be more prudent in preselecting securities for investors and make investment recommendations that are appropriate for investor risk profiles. **Investors** should also be mindful of the risks associated with financial products they are investing in and not just expected returns. Moreover, investors should heed the importance of financial statement analysis as well as risk diversification across different securities.

Going forward, the MPC would continue to closely monitor developments in the unrated bond market and other parts of the financial system for signs of underpricing of risks and other financial imbalances that could threaten financial stability. The MPC also viewed collaboration between various regulatory bodies and market participants to be increasingly important for effective risk monitoring and assessment, together with appropriate measures to manage risks that might arise in the future.

2.2 Outlook for the Thai economy

Under the Committee's assessment, Thailand's economic recovery was projected to gain further traction from 2016 and record growth of 3.4 percent in 2017 and 3.6 percent in 2018. Growth was expected to be driven by (1) the gradual recovery of exports of goods and services, supported by the global economic recovery and the swift recovery in tourism that was previously affected by measures to curb illegal tour operators, thanks in part to the extension of visa fee reduction/waiver period[7/], (2) fiscal impulse from public spending and investment in large infrastructure projects that would bolster private sector confidence and spur private investment in related areas, and (3) the gradual expansion in private consumption on the back of improving household income, declining debt burden from the first-car scheme, and low inflation that would sustain purchasing power of consumers.

Table 2.2 Summary of forecasts

Percent	2016*	2017	2018
GDP growth	3.2	3.4 (3.2)	3.6
Headline inflation	0.2	1.2 (1.5)	1.9
Core inflation	0.7	0.7 (0.8)	1.0

Note: * Outturn
() Monetary Policy Report December 2016
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

7/ Extension of the 1,000 baht reduction of visa-on-arrival fee and a visa fee waiver on tourists from 19 countries from February 2017 to August 2017.

Summary of key forecast assumptions

- Trading partners' economies were projected to continue recovering and record higher growth than previously assessed given stronger growth in major economies, especially the U.S., and a firmer recovery in Asian exports.
- The fed funds rate was raised at the FOMC meeting in March 2017, sooner than expected in the previous *Monetary Policy Report*. However, the expected number of policy rate hikes was still projected to total three times this year, and another three hikes next year.
- Asian currencies (excluding the Chinese yuan) were projected to remain close to the previous assessment, i.e., weakening in response to increases in the fed funds rate.
- The Dubai crude oil price was still expected to trend up but at a slower pace than the previous projection due to higher crude oil supply—particularly from the resumption of shale oil production in the U.S. as prices increased.
- Prices of non-energy commodities, particularly metals, were expected to be higher due to temporary supply disruptions as a result of disputes at major mining sites.
- Farm income was expected to improve on account of higher prices and output. Rice output, in particular, returned to normal levels after the drought subsided. Meanwhile, rubber prices increased in tandem with higher oil prices and increased demand from China. Farm income in 2018 was projected to continue increasing as weather conditions were expected to return to normal.
- Public spending at current price was projected to be higher than the previous assessment given that the fiscal budget for 2018 approved by the cabinet was larger than expected, and the details on the budget for the Pracharat project became clearer.

Table Summary of the main assumptions

Percent	2016*	2017	2018
Dubai crude oil price (U.S. dollar per barrel)	41.4	52.3 (53.5)	54.8
Non-oil commodity prices (%YoY)	-1.9	7.0 (1.2)	-0.3
Farm income (%YoY)	3.2	5.4 (1.9)	4.3
Public spending at current price (billion baht)[1/]	2,454	2,570 (2,547)	2,687
Public investment at current price (billion baht)[1/]	936	1,082 (1,089)	1,202
Fed funds rate (% at yearend)	0.63	1.38 (1.38)	2.13
Trading partners' economic growth rate (% YoY)[2/]	3.1	3.3 (3.1)	3.3
Regional exchange rate index (ex-China) per U.S. dollar[3/]	154.5	160.2 (160.3)	161.7

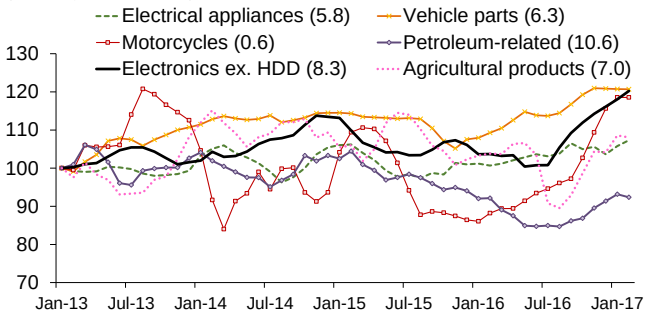
Notes: [1/] Including the spending assumption for the infrastructure investment project
[2/] Weighted by the share of Thai exports to each trading partner
[3/] Higher figures represent depreciation, lower figure represents appreciation
* Outturn
( ) Monetary Policy Report December 2016

Export of goods exhibited a stronger recovery across many product groups.

Chart 2.7 Export of goods exhibited a stronger recovery across many product groups, especially electronics

Value of exported goods

Seasonally adjusted index, 3-month moving average
(January 2013 = 100)



Note: Number indicated in () denote share to total exports in 2016
Source: Customs Department, calculation by Bank of Thailand.

The value of exported goods was expected to grow 2.2 percent in 2017 and 2.0 percent in 2018. The firmer global economic recovery had a positive impact on Thai exports which began to show signs of recovery since the fourth quarter of 2016. Exports of electronics and electrical components, in particular, benefited from the increasing popularity of smart phones and Internet of Things (IoT) products in foreign markets. At the same time, the relocation of production sites to Thailand by some manufacturers contributed to additional production capacity for products such as car tires and solar cells. Exports of food and agricultural goods also began to improve as the shortage of domestic raw materials eased. Aside from the more positive outlook for export volume, export prices were projected to increase in tandem with crude oil prices, especially commodities such as rubber and oil-related goods such as petroleum products.

Nevertheless, Thailand's export outlook remained constrained by shifts in the global trade structure as well as some difficulties faced by domestic producers that might take time to resolve, such as the declining consumer demand for hard disk drives and the loss of competitiveness in textile and garment production. Moreover, in the period ahead, there remained risks stemming from potential changes in the U.S. trade policies that could affect global trade, and other downside risks that could contribute to slower-than-expected global growth.

Exports of services swiftly recovered and were expected to continue expanding.

Exports of services were expected to record strong growth on the back of tourism. The number of foreign tourists recovered rapidly from the impact of the curb on illegal tour operators in the previous period (Box 3: *Thailand's tourism outlook*), partly thanks to the extension of the visa fee reduction/waiver which helped eased expenses for foreign tourists. While measures to curb illegal tour operators negatively impacted the growth of Chinese tourists traveling on group tours over the short run, these measures would help raise the standards and quality of tourism-related services and benefit Thailand's tourism in the longer term. In addition, the limit on sales of tour packages to Korea imposed by the Chinese authorities on Chinese tourists was expected to induce some Chinese tourists to visit Thailand instead. In this light, **the Committee raised the estimates of foreign tourists to 34.5 and 37.1 million in 2017 and 2018, respectively.**

Given the firmer recovery in the exports of goods, growing exports of services, and higher oil prices, albeit lower than past projections, **Thailand's current account was projected to record a surplus of 36.9 and 33.1 billion U.S. dollar in 2017 and 2018, respectively.**

Thailand's tourism outlook

Thailand's exports of tourism services continued to expand in 2016 and remained an important driver of growth. The tourism industry faced some obstacles in the fourth quarter of 2016 due to the government's effort to tackle illegal tours as well as the reduced entertainment activities. Yet the industry's overall growth remained high. Revenues from tourism made up 12.3% of Thailand's GDP with a large share of income coming from European and Chinese tourists (Charts 1 and 2).

For 2017, tourism receipts are projected to expand in tandem with the global economic recovery, supported by greater traveling convenience as well as declining travel costs due to the expansion of low-cost airlines routes. Revenues from exports were expected to reach 12.8% of GDP this year. The number of foreign tourists was also projected to grow 5.8%, faster than the global average (Chart 3), which reflected an increasing market share for Thailand in global tourism.

Going forward, the Thai government plans to tap into other potential markets to improve both the quantity and quality of tourists by:

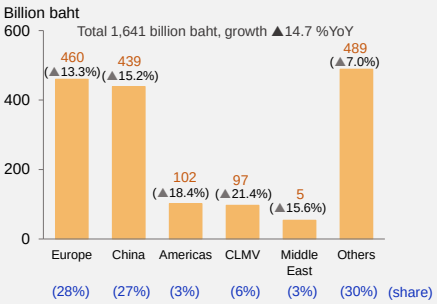
(1) Raising the number of inbound tourists, especially from CLMV

The government is pursuing collaborative measures with CLMV governments, such as signing a memorandum of understanding (MOU) to support cross-border tourism and co-organizing tourism promotion programs along with improving transportation connectivity within the region under the "Five Countries, One Destination" plan.

(2) Attracting tourists with high spending per trip to increase tourism revenues

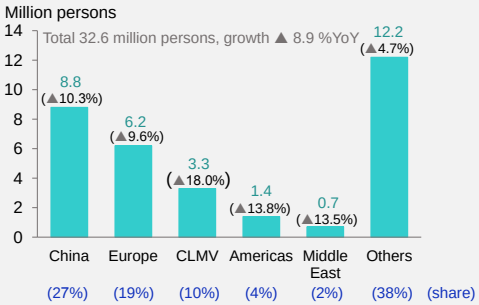
Tourists from the Americas and the Middle East, for instance, spend as much as 73,000 baht per person per trip (average spending by foreign tourists is 50,364 baht/person/trip). Strategies deployed to attract this group include partnering with major airlines from the Americas and the Middle East. **Another strategy involves attracting Chinese tourists, which still have high growth potential,** through visa fee waivers and the promotion of Thailand tourism in China. Nevertheless, the government's ability to successfully restructure the tourism industry toward quality growth and its progress in improving infrastructure and public services so that tourism could be a sustainable driver of growth remains to be seen.

Chart 1 A majority of tourism receipt came from European and Chinese tourists.
Tourism revenue in 2016



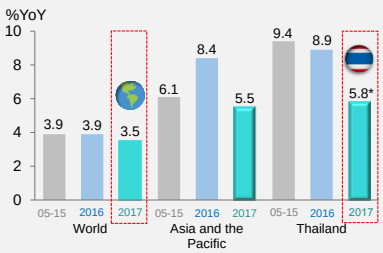
Note: ▲ denote growth compared to the previous year
Source: Department of Tourism

Chart 2 Chinese tourists made up the largest share of all international arrivals.
International arrivals in 2016



Note: ▲ denote growth compared with the previous year
Source: Department of Tourism

Chart 3 Tourist arrivals in Thailand was projected to grow.
Projected tourist arrivals by destination in 2017



Note: *Projected by Bank of Thailand
Source: UNWTO (January 2017) and the Bank of Thailand

Public spending was projected to pick up.

Public spending was expected to remain a key growth driver throughout the forecast horizon, attributable mainly to investment by the central government and state-owned enterprises. Public spending was expected to be higher than projected in the previous *Monetary Policy Report* due to (1) greater clarity in the details of the Pracharat projects, (2) higher-than-expected fiscal budget for 2018 approved by the cabinet, and (3) faster budget disbursement for the telecommunication infrastructures upgrade project [8/]. However, investment by state-owned enterprises was slower than expected due to the partial rescheduling of this year's dual-track rail construction to next year as a result of the procurement process review. In addition, risk to the public spending forecast remained tilted to the downside, as the implementation of public investment projects potentially faced longer-than-expected delays.

Private consumption was expected to recover at a gradual pace.

Private consumption was projected to gradually expand in the period ahead driven by (1) improved farm income due to higher rubber prices that resulted from higher oil prices and increased demand from China, and also due to higher rice output gains as the weather conditions returned to normal, (2) robust income growth for households in the service sector supported by strong growth in services and tourism, and (3) declining debt burden from the first-car scheme that would help households' purchasing power somewhat. Nevertheless, in the short term, income and employment in the manufacturing sector did not benefit much from the recovery of the exports of goods as the recovery was mostly related to products manufactured in capital-intensive industries.

Private investment showed signs of recovery in certain business sectors.

Private investment was projected to recover going forward, but growth would likely remain low. There were signs of recovery emerging in certain business sectors such as telecommunications and electronics that benefited from growing demand for related products, and in alternative energy businesses that received investment benefits from the BOI. In the near term, private investment might not benefit much from the recovery in private consumption and merchandise exports, as production capacity remained ample. In the period ahead, however, private investment would likely pick up as private consumption and exports would gain stronger traction. Government policies would also play a critical role in supporting private investment through public-private partnership (PPP), plans to expedite infrastructure investment, and industrial estate developments in the Eastern Economic Corridor (EEC), all of which would help bolster business confidence and attract foreign investment to Thailand.

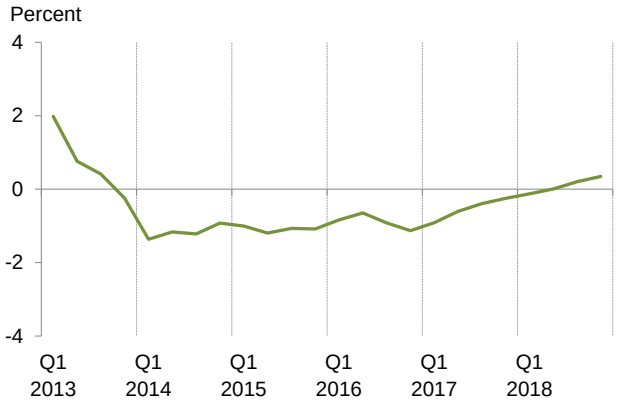
Inflation was expected to gradually rise.

In the period ahead, inflationary pressures were projected to increase from both demand- and supply-side factors. Demand-pull inflationary pressures would increase gradually in tandem with the economic recovery, as reflected in the closing output gap. Cost-push pressures in 2017 would rise on account of a strong rebound in oil and commodity prices

[8/] National digital infrastructure development project on hi-speed internet network expansion using optical fiber cables

from a low base in the previous year, despite being partially offset by the base effect of higher fresh food prices during the drought in 2016. Upward cost pressures were expected in 2018 because of projected increases in fresh food prices as conditions return to normal and also the gradual increase in oil prices.

Chart 2.8 Output Gap

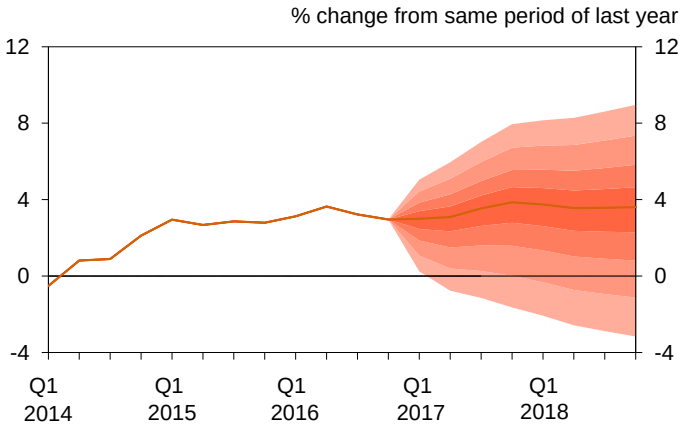


The Committee viewed that headline inflation would rise and gradually approach the midpoint of the target. **Core inflation was projected to average at 0.7 percent in 2017 and 1.0 percent in 2018 and headline inflation was projected to average at 1.2 percent in 2017 and 1.9 percent in 2018.**

Growth and inflation forecasts faced more downside risks than upside risks.

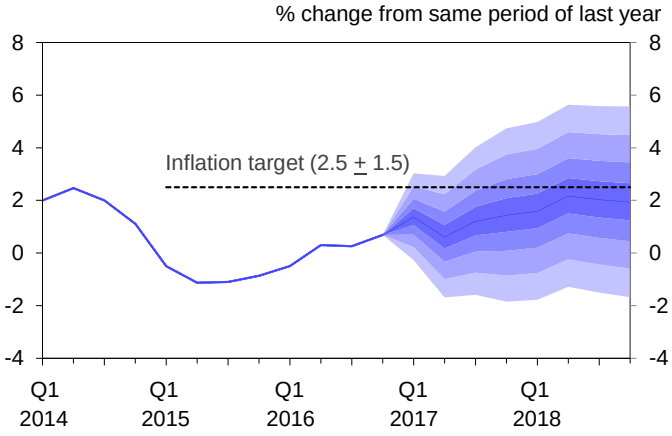
The Committee assessed that there were more downside risks than upside risks to the growth projection, and the overall balance of risks tilted further to the downside compared with the previous assessment. Such assessment reflected the growth forecast fan chart that remained skewed to the downside throughout the forecast horizon and to a greater extent than in the previous *Monetary Policy Report*. An important downside risk was the uncertainty surrounding the U.S. trade policies that could affect the global economic outlook and lead to a significant slowdown in Thailand's trading partner economies. In addition, public investment might face delays especially in larger investment projects. Meanwhile, financial stability concerns in China continued to warrant close monitoring. On the other hand, there were some upside risks to the baseline growth projections. Public investment might crowd in more private investment than in the baseline scenario. The U.S. economy might achieve stronger growth from its domestic stimulus policies. In line with risks to growth, the Committee viewed the forecasts for headline and core inflation to be subject to more downside risks than upside risks.

Chart 2.9 Growth forecasts



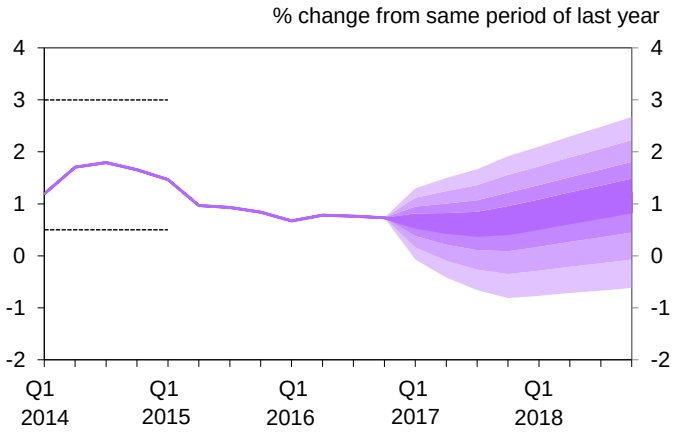
Note: Fan chart covers 90 percent of probability distribution

Chart 2.10 Headline inflation forecasts



Note: Fan chart covers 90 percent of probability distribution

Chart 2.11 Core inflation forecasts



Note: Fan chart covers 90 percent of probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2016*	2017	2018
GDP growth	3.2	3.4 (3.2)	3.6
Domestic demand	2.7	3.2 (3.2)	3.3
Private consumption	3.1	2.7 (2.6)	3.1
Private investment	0.4	2.4 (1.6)	3.7
Government consumption	1.6	2.2 (3.2)	1.9
Public investment	9.9	11.8 (11.9)	7.5
Volume of exported goods and services	2.1	2.1 (0.6)	2.8
Volume of imported goods and services	-1.4	2.6 (2.4)	2.9
Current account (billion, U.S. dollars)	46.4	36.9 (26.9)	33.1
Value of exported goods	0.0	2.2 (0.0)	2.0
Value of imported goods	-4.7	7.2 (7.8)	5.3

Note: *Outturns

( ) Monetary Policy Report December 2016

3. Monetary Policy

The Committee viewed the Thai economy to have improved and would expand at a pace faster than previously assessed.

The Committee viewed that the Thai economic outlook had improved on the back of a stronger recovery of merchandise exports. Tourism quickly recovered after having been impacted by the government's regulation on illegal tours. However, the spillovers of improving exports on employment and household income remained limited. Meanwhile, private investment still concentrated in some export-oriented businesses. As a result, domestic demand would recover only at a gradual pace. The Committee viewed that the Thai economy would face considerable uncertainties on the external front, particularly those pertaining to international trade policy in the U.S. that remain subject to uncertainty, together with potential responses by its major trading partners. These could undermine Thai exports in the period ahead. The Committee would closely monitor developments that might affect the continuation of and risks to the Thai economic recovery in order to formulate appropriate monetary policy.

The Committee projected inflation to be lower than the projection in the previous *Monetary Policy Report* due to lower-than-expected outturns and the downward revision of oil price projection. Headline inflation already returned to the lower bound of the target and was expected to gradually rise. Nevertheless, headline inflation could fluctuate in the near term due to the base effect caused by elevated fresh food prices during the drought last year. Also, there were more downside risks in connected with variations in global oil prices given the uncertainties of global oil supply. Core inflation would stay at a low level, reflecting low demand-pull inflationary pressures due to the slowly recovering domestic demand. Meanwhile, the public's medium-term inflation expectations remained close to the midpoint of the target.

Overall financial conditions remained accommodative. Total financing increased for businesses that were well recovering. Financing costs edged up somewhat but would not significantly affect corporate financing.

The Committee deemed overall financial conditions accommodative and conducive to economic growth with ample liquidity in the financial system and low real interest rates. Commercial bank loans grew at a low rate, with growth concentrated in loans extended to large corporates and SMEs in manufacturing and trade sectors with a recovery outlook. The low credit growth was partly a result of falling loan demand by businesses and households. At the same time, banks remained cautious in lending, especially to SMEs and households, given the outlook for nonperforming loans. Nonetheless, the Committee assessed that overall financing costs of the private sector through credit markets continually declined according to the analysis based on the new loan rate (Box: *An analysis of monetary policy transmission based on the new loan rate*). Bond financing increased but still concentrated in some businesses. However, financing conditions in the bond markets tightened somewhat as external factors raised Thailand's long-term government bond yields (with maturity over five years). However, the Committee assessed that the impact on the private sector was limited. Most businesses in Thailand that raise funds in the bond markets usually issue short- to medium-term bonds (with maturity of 1-3 years); theirs yields currently did not rise much and still remained below historical averages. In addition, the recent defaults of some corporate bonds issued by unrated companies ("unrated bonds") caused some difficulty to other

businesses in rolling over the bonds of this type. Nonetheless, this had not significantly affected overall financing abilities of the private sector in the bond markets.

Regarding the exchange rates, the Committee regarded uncertainties surrounding economic policy in the U.S. and political developments in Europe as factors making the baht appreciate against currencies of major trading partners and competitors. This was due to Thailand's relatively stronger fundamentals and external positions. In the period ahead, capital flows and exchange rates would face heightened volatility stemming from uncertainties on the timing and pace of policy rate hikes, especially by the Fed, and on monetary policy directions of major advanced economies. Therefore, appropriate foreign exchange risk management by the private sector would be critical going forward.

Financial stability remained sound. However, there were pockets of risks that warranted close monitoring.

Under the Committee's view, overall financial system remained sound with sufficient cushion against economic and financial volatilities on both domestic and external fronts. However, there remained pockets of risks that warranted close monitoring such as the deterioration in SME and household debt repayment ability and yield-searching behavior that might lead to underpricing of risk. The Committee would closely follow these developments going forward. Moreover, some committee members viewed that risks to financial stability could rise in the future as a result of the low-for-long interest rate environment. Increased long-term bond yields over the recent period also led firms to raise funds through shorter-term instruments. This could lead to accumulation of risks in the financial system. Consequently, collaboration among relevant regulatory authorities would play a greater role in reducing risks and vulnerability in Thailand's financial system.

The Committee voted unanimously to maintain the policy rate at 1.50 percent to allow monetary conditions to be accommodative and conducive to continuation of economic growth.

The Committee voted unanimously to maintain the policy rate at 1.50 percent at the meetings on February 8 and March 29. In deliberating their policy decision, the Committee assessed that Thailand's growth outlook improved and headline inflation was expected to rise and approach the midpoint of the target. However, considerable risks on the external front remained, particularly those pertaining to the fragile global economic recovery and U.S. trade policy which might restrain the recovery of Thai exports. Moreover, uncertainties regarding monetary policy directions of major advanced economies could induce further capital flow and exchange rate volatilities. Thus, the Committee viewed that monetary policy should remain accommodative until economic recovery is more robust and stronger. This would also support inflation to return to the midpoint of the target. The Committee would stand ready to utilize an appropriate mix of available policy tools in order to ensure that monetary conditions are conducive to economic growth, while ensuring financial stability.

An analysis of monetary policy transmission based on the new loan rate

Understanding how the policy interest rate affects retail loan rates is key to measuring the effectiveness of monetary policy transmission. In the past, the Bank of Thailand used two key retail interest rates to evaluate the impact of monetary policy transmission: (1) the minimum lending rate (MLR), which is the rate at which commercial banks charge on high-quality and large-size customers, and (2) the effective lending rate (ELR), which is based on interest payments that banks receive from the borrowers. However, both MLR and ELR have limitations.

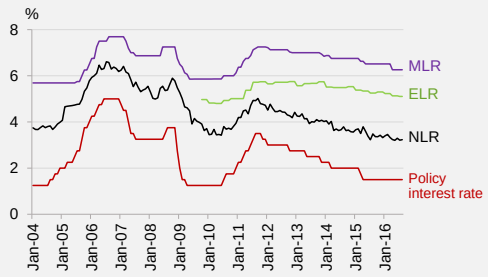
(1) The MLR is a posted rate determined by commercial banks. Although it generally moves in the same direction as the policy rate, actual interest rates on loans might be different than the MLR depending on credit risk premium that banks assign particular borrowers. According to the data from commercial banks, many customers with good financial position are charged at rates lower than the MLR.

(2) The ELR is based on interest payments that commercial banks receive on all loans. This measure takes into account (a) existing loan contracts that have not expired, (b) new loans, and (c) some loan contracts that were already default. Consequently, it does not reflect the impact of policy rate changes on the interest rates applied on new loans.

Based on analyses using the "new loan rate" (NLR)[9/], calculated from a large database of loan arrangements (LAR)[10/] with high granularity, **transmission of policy interest rates to retail rates has been more than previously estimated using only the MLR or ELR.** In particular, the MLR and ELR are found to adjust only slowly relative to the policy rate given recent interest rate cuts. Meanwhile, the NLR has been more responsive to the change in the policy interest rate (Chart 1). In addition, a regression[11/] analysis reveals that, despite a slower response of the MLR and NLR to changes in the policy rate recently, the NLR is more responsive than the MLR at any given point in time (Chart 2). At present, a cut of policy interest rate by 1 percentage point corresponds to a drop of 0.46 percentage point in the NLR and a decline of only 0.33 percentage point in the MLR.

Chart 1 NLR is more responsive to policy interest rate movement than the MLR and ELR

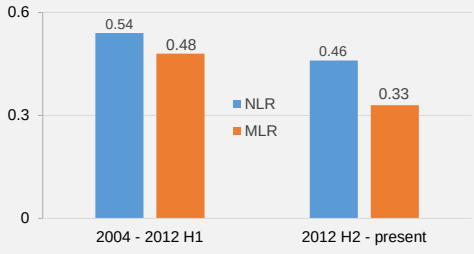
Policy interest rate, MLR, ELR, NLR



Source: The Bank of Thailand

Chart 2 NLR responses more quickly to the policy interest rate than the MLR at any given point in time

Changes of interest rates to changes in the policy rate (ΔR /ΔRP)



Source: The Bank of Thailand

[9/] The NLR is calculated from interest payments on loan contracts by 14 Thai commercial banks excluding loans to households and financial intermediaries and is weighted by loan size.

[10/] LAR is a database of loans at the loan contract level. The Bank of Thailand requires financial institutions to report the information at the end of every month since January 2004. Such financial institutions include (1) Thai commercial banks, subsidiaries of foreign commercial banks, and branches of foreign commercial banks with outstanding loans exceeding 20 million baht, (2) retail commercial banks and finance companies with outstanding loans exceeding than 5 million baht, (3) credit fonciers with outstanding loans exceeding than 1 million baht. The LAR database currently covers 70% of total outstanding loans by commercial banks, which should be adequate for the study of loans in Thailand.

[11/] The sensitivity of movement of interest rates to changes in the policy interest rate is estimated from the following regression: $\Delta R = b_0 + b_1 \Delta RP + b_2 \Delta RP_{t-1} + b_3 \Delta RP_{t-2}$ where *R* denotes MLR or NLR and *RP* represents the policy interest rate. The response of MLR or NLR to the change in policy interest rate is calculated from *b1+b2+b3*. The study divides the analysis into two periods: (1) from 2003 to the first half of 2012, and (2) from the second half of 2012 to present. The results suggest a structural change in monetary policy transmission after the Great Flood.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2014	2015	2016			
			Q1	Q2	Q3	Q4
GDP growth	2.9	3.2	3.1	3.6	3.2	3.0
Production						
Agriculture	-5.7	0.6	-2.0	-0.4	0.9	3.2
Non-agriculture	3.9	3.5	3.7	3.9	3.2	3.1
Manufacturing	1.5	1.4	-0.2	2.2	1.6	2.1
Construction	17.0	8.3	11.9	9.9	5.2	6.1
Wholesales and retail trade	3.9	5.0	4.6	4.7	5.2	5.6
Hotels and restaurants	14.6	10.3	12.3	10.8	13.5	4.8
Transport, storage, and communication	5.1	5.5	6.4	4.2	6.5	5.1
Financial intermediation	8.8	6.2	5.9	6.2	5.8	6.7
Real estate, renting, and business activities	1.9	1.8	2.2	2.1	1.1	1.8
Expenditure						
Domestic demand	2.9	2.7	4.4	3.5	0.9	2.2
Private consumption	2.2	3.1	2.8	4.0	3.0	2.5
Private investment	-2.2	0.4	2.4	0.3	-0.8	-0.4
Government consumption	3.0	1.6	9.1	2.4	-5.2	1.5
Public investment	29.3	9.9	12.9	12.8	5.8	8.6
Imports of goods and services	0.0	-1.4	-5.2	-2.5	-1.1	3.4
imports of goods	0.2	-2.1	-7.6	-2.8	-1.5	3.6
imports of services	-1.0	1.7	5.7	-1.3	0.5	2.1
Exports of goods and services	0.7	2.1	4.8	1.0	1.4	1.1
exports of goods	-3.4	0.0	0.8	-1.9	-0.4	1.4
exports of services	17.1	9.3	18.1	11.6	7.7	0.4
Trade balance (million, U.S. dollars)	26,841	35,752	11,839	8,307	8,973	6,633
Current account (million, U.S. dollars)	32,149	46,412	17,852	8,391	10,380	9,789
Financial account (million, U.S. dollars)	-17,102	-24,804	-300	-3,062	-9,096	-12,347
International reserves (billion, U.S. dollars)	157	172	175	179	180	172
Unemployment rate (%)	1.0	1.0	1.0	1.1	0.9	1.0
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	0.9	1.0	1.1	1.1

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2015	2016	2016				2017	
			Q1	Q2	Q3	Q4	Jan	Feb
1. Financial market sector								
Bond market								
Bond spread (10 years - 2 years)	1.1	0.6	0.7	0.5	0.6	0.8	1.1	1.1
Equity market								
SET index (end of period)	1,288.0	1,542.9	1,407.7	1,445.0	1,483.2	1,542.9	1,577.3	1,559.6
Actual volatility of SET index[1/]	13.9	15.2	15.9	11.9	13.6	15.2	7.9	6.6
Price to Earnings ratio (P/E ratio) (times)	22.6	18.6	20.7	22.0	21.3	18.6	19.1	18.1
Exchange rate market								
Actual volatility of Thai baht (%annualized)[2/]	5.1	4.4	4.4	4.6	3.8	5.0	3.9	3.1
Nominal Effective Exchange Rate (NEER)	108.5	106.2	106.7	105.1	105.8	107.0	108.3	108.7
Real Effective Exchange Rate (REER)	104.4	100.5	100.8	100.0	100.3	100.7	101.8	n.a
2. Financial institution sector[3/]								
Minimum Lending Rate (MLR)[4/]	6.50	6.26	6.51	6.26	6.26	6.26	6.26	6.26
12-month fixed deposit rate[4/]	1.40	1.38	1.40	1.38	1.38	1.38	1.38	1.38
Capital adequacy								
Capital funds / Risk-weighted asset (%)	17.4	18.0	17.5	17.5	18.5	18.0	n.a	n.a
Earning and profitability								
Net profit (billion, Thai baht)	192.3	199.2	51.2	50.9	49.8	47.4	n.a	n.a
Return on assets (ROA) (times)	0.9	1.1	1.2	1.2	1.1	1.1	n.a	n.a
Liquidity								
Loan to Deposit and B/E (%)	97.0	96.3	94.8	96.4	96.9	96.3	n.a	n.a
3. Household sector								
Household debt to GDP (%)	81.2	n.a	80.6	80.3	80.0	79.9	n.a	n.a
Financial assets to debt (times)	2.6	n.a	2.5	2.6	2.6	n.a	n.a	n.a
Non-Performing Loans (NPLs) of commercial banks (%)								
Consumer loans	2.6	2.7	2.6	2.6	2.7	2.7	n.a	n.a
Housing loans	2.4	2.9	2.6	2.7	2.8	2.9	n.a	n.a
Auto leasing	2.3	1.8	2.1	1.9	1.8	1.8	n.a	n.a
Credit cards	4.0	3.7	4.7	4.2	5.1	3.7	n.a	n.a
Other personal loans	2.7	2.9	2.7	2.7	2.9	2.9	n.a	n.a
4. Non-financial corporate sector[5/]								
Operating profit margin (OPM) (%)	7.5	8.3	8.5	8.7	8.2	7.8	n.a	n.a
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	n.a	n.a
Interest coverage ratio (ICR) (times)	5.5	6.5	6.5	7.1	5.8	6.8	n.a	n.a
Current ratio (times)	1.7	1.7	1.7	1.7	1.7	1.6	n.a	n.a
Non-Performing Loans (NPLs) of commercial banks (%)								
Large businesses	1.6	1.5	1.7	1.9	1.9	1.5	n.a	n.a
SMEs	3.5	4.3	3.7	3.8	4.0	4.3	n.a	n.a

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2015	2016	2016				2017	
			Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector								
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)								
Total	59,667	61,452	15,487	15,611	14,149	16,205	2,534	3,572
Single-detached and semi-detached houses	13,152	13,409	3,511	3,503	3,216	3,179	538	821
Townhouses and commercial buildings	19,210	20,187	5,092	5,410	4,718	4,967	873	1,317
Condominiums	27,305	27,856	6,884	6,698	6,215	8,059	1,123	1,434
Number of new housing units launched for sale (Bangkok and Vicinity) (units)								
Total	107,988	110,575	28,426	19,680	31,017	31,452	4,168	8,386
Single-detached and semi-detached houses	17,637	19,433	5,033	3,099	6,328	4,973	145	1,048
Townhouses and commercial buildings	27,518	32,792	7,591	6,702	10,638	7,861	1,005	5,304
Condominiums	62,833	58,350	15,802	9,879	14,051	18,618	3,018	2,034
Housing price index (2009 = 100)								
Single-detached houses (including land)	129.0	130.8	129.4	133.7	131.3	128.8	129.2	128.1
Townhouses (including land)	135.9	137.6	138.8	139.1	136.8	135.6	136.7	137.7
Condominiums	158.7	166.2	160.0	162.1	169.2	173.6	170.6	172.3
Land	157.2	171.2	169.5	173.7	170.2	171.3	172.0	172.5
6. Fiscal sector								
Public debt to GDP (%)	44.4	42.2	44.0	42.8	42.8	42.2	42.0	n.a
7. External sector								
Current account balance to GDP (%)[6/]	8.1	11.4	17.9	8.4	10.1	9.4	n.a.	n.a.
External debt to GDP (%)[7/]	32.1	32.5	34.3	35.4	34.5	32.5	n.a.	n.a.
External debt (billion, U.S. dollars)	131.4	131.5	139.1	143.1	139.3	131.5	133.3	135.0
Short-term (%)	40.0	40.3	42.3	42.5	40.2	40.3	40.5	40.8
Long-term (%)	60.0	59.7	57.7	57.5	59.8	59.7	59.5	59.2
International reserves / Short-term external debt (time	3.0	3.2	3.0	2.9	3.2	3.2	3.3	3.3

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2017				2018			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
10-12	0	0	0	1	1	1	2	2
8-10	0	0	2	4	4	4	5	6
6-8	1	4	9	13	13	12	12	13
4-6	18	22	24	24	22	21	20	19
2-4	49	36	30	25	24	23	22	21
0-2	28	26	21	19	19	19	18	17
(-2)-0	5	10	10	10	11	12	12	12
< (-2)	0	2	3	5	6	8	9	10

Table: Probability distribution of core inflation forecast

Percent	2017				2018			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	1	1	2
2.5-3.0	0	0	0	1	1	2	3	5
2.0-2.5	0	0	1	3	5	7	8	10
1.5-2.0	1	4	6	9	11	14	15	16
1.0-1.5	17	18	18	18	19	20	20	19
0.5-1.0	44	32	27	23	22	21	19	18
0.0-0.5	30	27	24	21	19	17	15	14
(-1)-0.0	7	14	15	14	12	11	10	9
(-2)-(-1)	1	4	6	7	6	6	5	4
< -2	0	1	2	3	3	2	2	2

Table: Probability distribution of headline inflation forecast

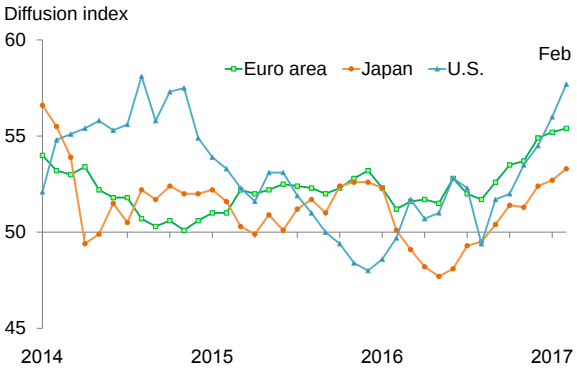
Percent	2017				2018			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 7	0	0	0	0	0	1	1	1
6-7	0	0	0	0	1	2	1	1
5-6	0	0	1	2	2	4	4	4
4-5	0	0	3	5	6	9	9	8
3-4	3	2	8	11	12	15	15	14
2-3	20	10	17	17	18	19	19	18
1-2	39	23	24	21	20	19	19	18
0-1	28	29	22	19	18	14	15	15
(-1)-(0)	8	22	15	13	12	9	10	10
(-2)-(-1)	1	10	7	7	7	5	5	6
(-3)-(-2)	0	3	3	3	3	2	2	3
(-4)-(-3)	0	1	1	1	1	1	1	1
< (-4)	0	0	0	0	0	0	0	0

4.2 Data pack

The Global Economy

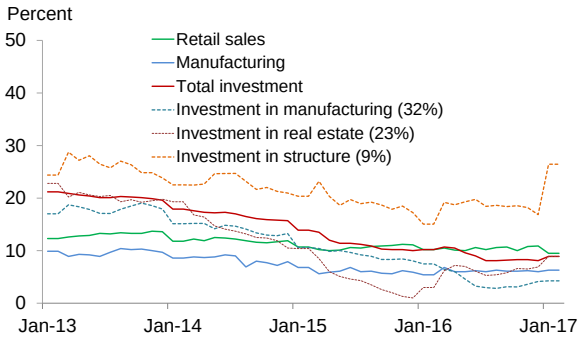
Advanced economies continued to recover on the back of gradual improvements in employment and domestic demand. China's infrastructure investment increased but the overall growth could slow down in the period ahead due to the government's economic reform. Asian exports steadily improved, especially exports of electronics.

Manufacturing Purchasing Manager Index



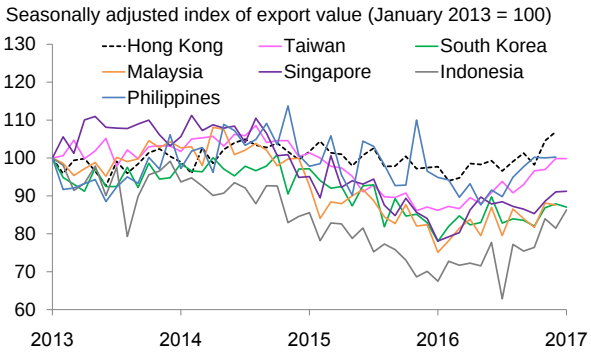
Source: Bloomberg and Eurostat

China's economic indicators (Change from the same period last year)



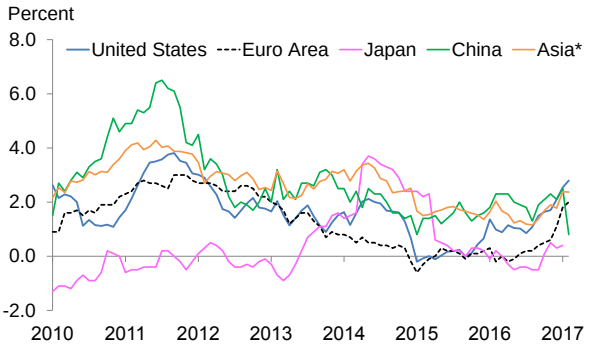
Note: () denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

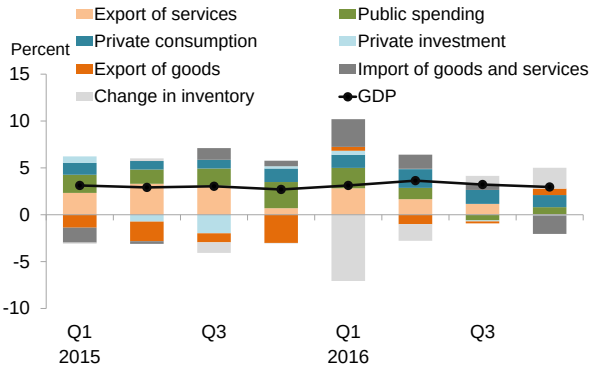


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

The Thai economy

In the fourth quarter of 2016, the Thai economy continually recovered on the back of private consumption, public spending, and merchandise exports that rebounded and showed a more broad-based recovery. The number of international arrivals quickly recovered and would likely continue increasing.

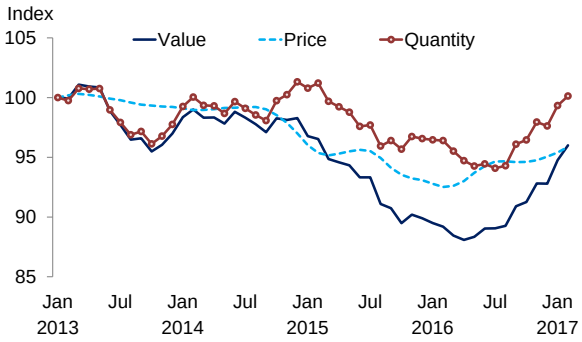
Contribution to Thailand's GDP growth[1/]



Note: [1/] Calculated by Chain Volume Measure method (CVM)
Source: Office of National Economic and Social Development Board, calculations by Bank of Thailand.

Thai exports (excluding gold): value, price and quantity

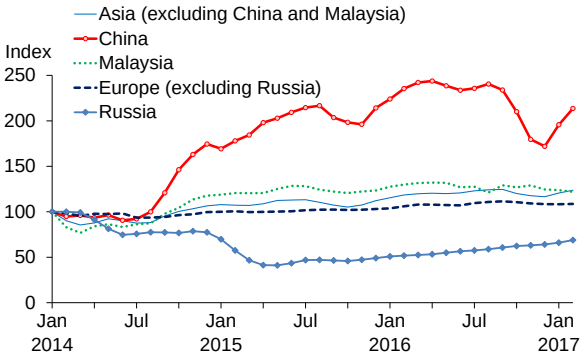
(3-month moving average, sessionally adjusted; January 2013 = 100)



Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

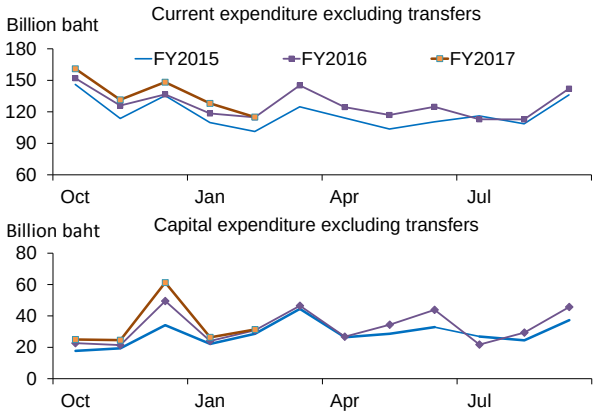
Index of foreign tourists classified by nationality

(3-month moving average, sessionally adjusted; January 2014 = 100)



Source: Department of Tourism

Public spending by central government

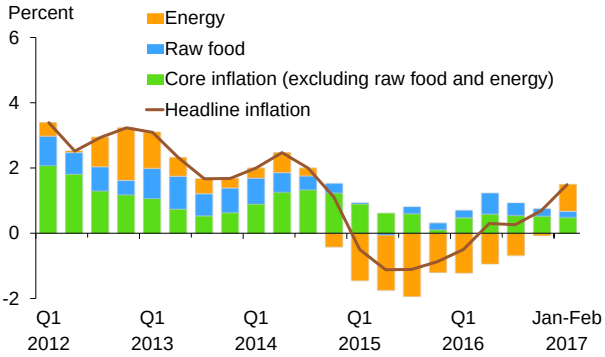


Source: Bureau of Budget, Fiscal Policy Office

Inflation

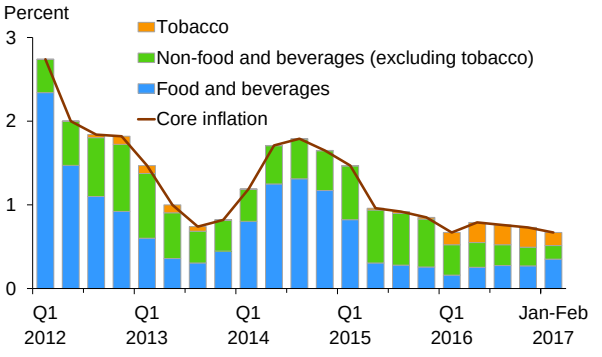
Headline inflation increased and reached the lower bound of the inflation target due mainly to higher energy prices. Meanwhile, core inflation slightly declined on account of the high base due to tobacco prices. Medium-term inflation expectations by professional economists stayed close to the midpoint of the target.

Contribution to headline inflation



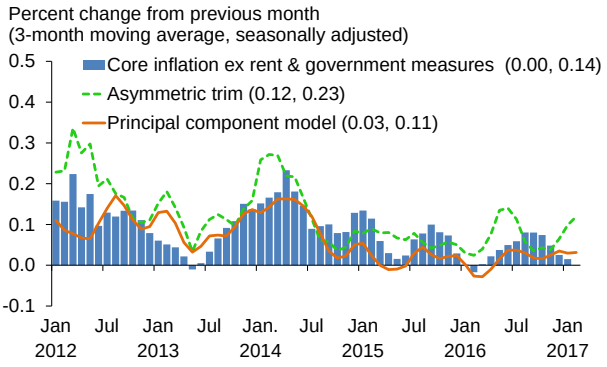
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

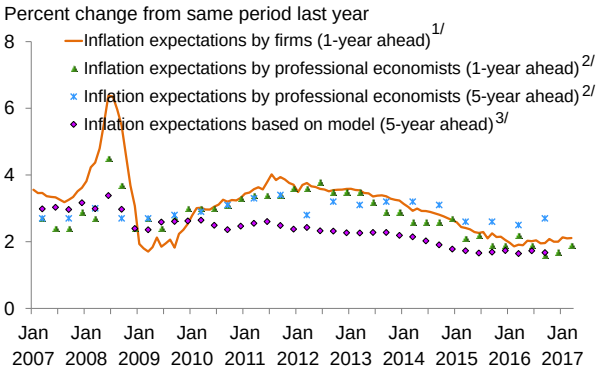
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of November 2016, while the second value is 2004 - 2014 average; Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; Principal component model calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

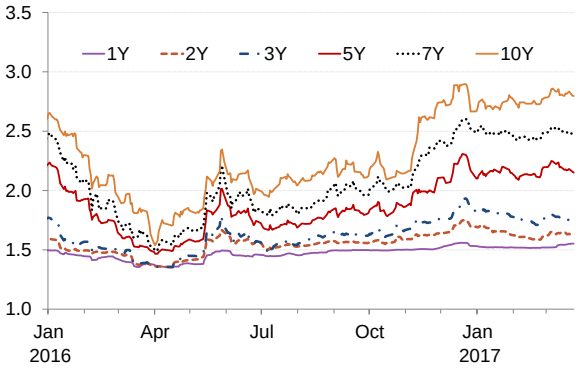


Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial and monetary conditions

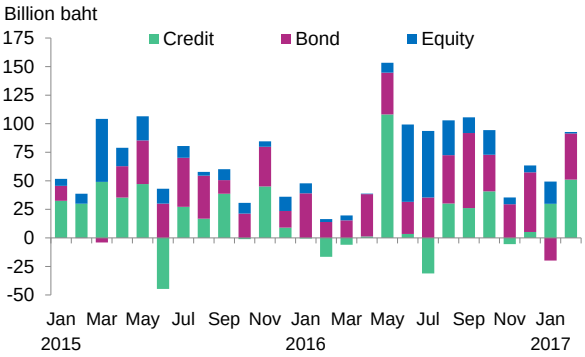
Government bond yields increased due to global factors. Corporate credits showed signs of higher growth, especially loans extended to export-related businesses. The Thai baht appreciated against the U.S. dollar, consistent with movements of regional currencies.

Thai government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

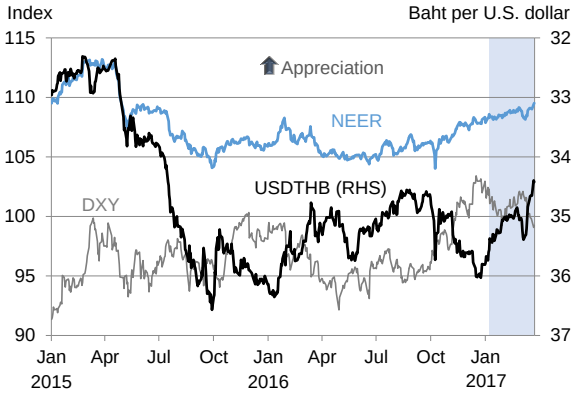
Total corporate financing by instruments*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bond excluding commercial banks, and newly issued equities.

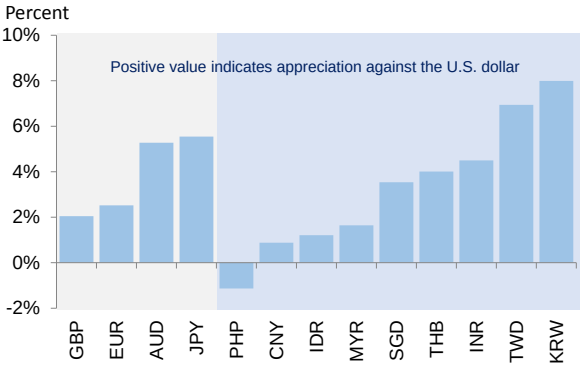
Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 28 March 2017)

Currency movements vis-a-vis the U.S. dollar (28 March 2017 compared to end-2016)

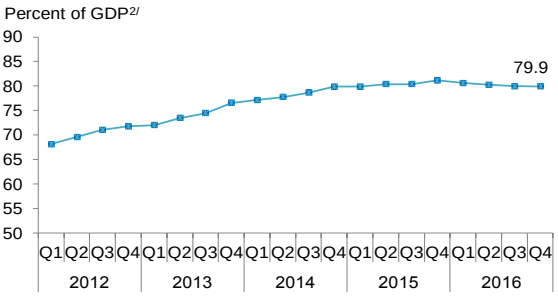


Source: Bank of Thailand and Reuters (data as of 28 March 2017)

Stability: household sector

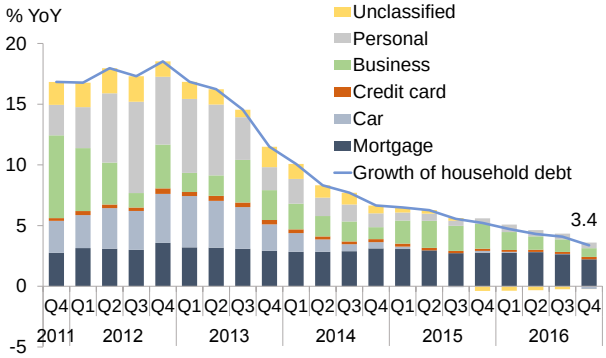
The ratio of household debt to GDP continued to trend down recently albeit slowly. However, deterioration in debt repayment ability warranted close monitoring.

Household debt[1/]



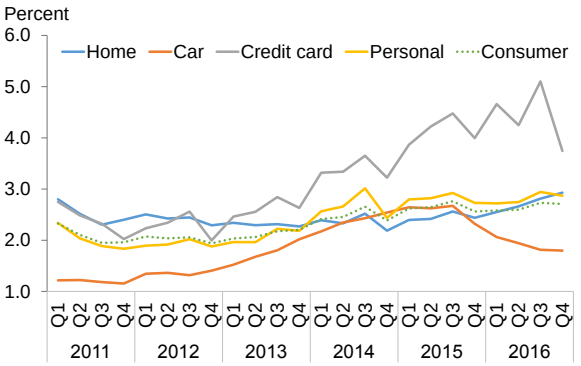
Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
Source: Bank of Thailand

Contribution to growth of household debt



Note: [1/] Loans given to households by financial institutions
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

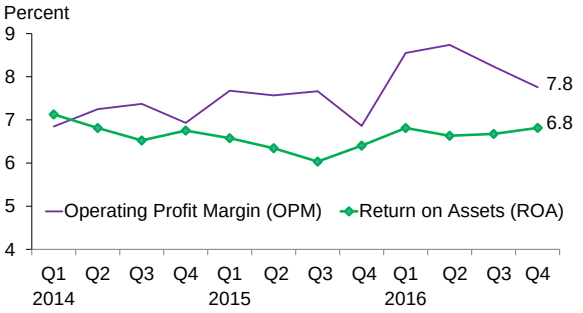


Source: Bank of Thailand

Stability: corporate sector

Overall stability in the corporate sector remained sound. However, small enterprises had yet to benefit from the economic recovery, with NPLs still rising.

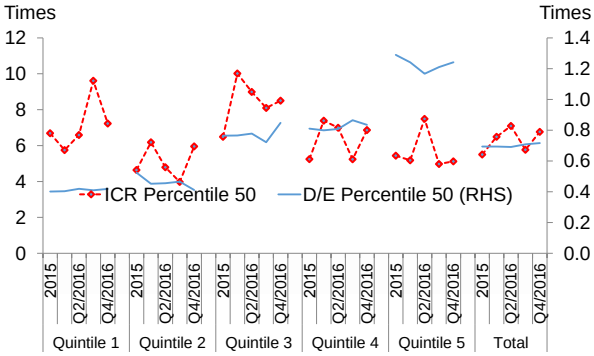
Operating Profit Margin (OPM) and Return on Assets (ROA)*



Note: * Median estimates; ROA is returns to average assets. OPM is operating profits to total sales.

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

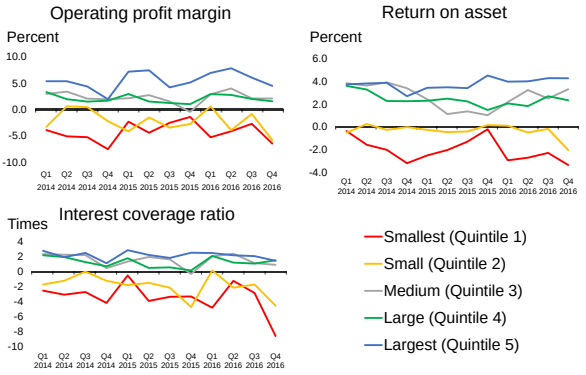
Interest Coverage Ratio (ICR) and Debt to Equity (D/E) ratios, classified by firm size



Note: Median estimates; Firms divided based on their asset size into 5 quintiles from smallest to largest. Firms with ICR lower than 1.5 and D/E over 2 warrant a close monitoring.

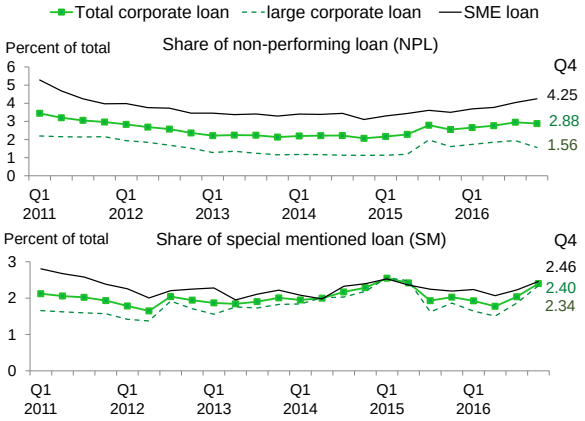
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Profitability and debt serviceability at 25th percentile of each group of firm size



Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

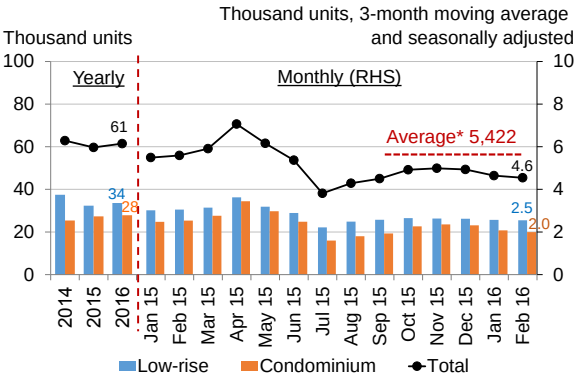


Source: Bank of Thailand

Stability: real estate sector

Demand for residences steadied after the end of real estate stimulus measures. Growth of the property sector slowed down as excess supply in some areas remained high. Condominium prices grew at a slower rate from last year, which was in line with land prices.

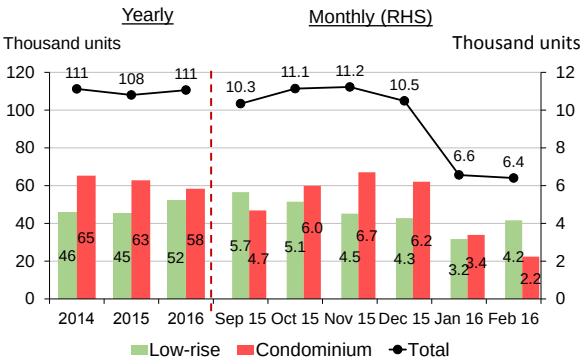
Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



Note: *Average during 2010-2016, excluding periods with government's stimulus measures (November 2015-April 2016)

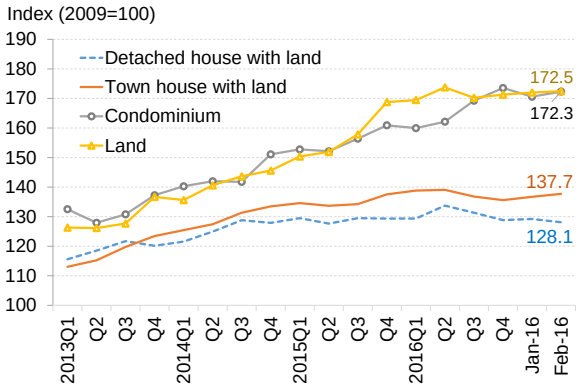
Source: Bank of Thailand

New residential projects launched in Bangkok and its vicinity



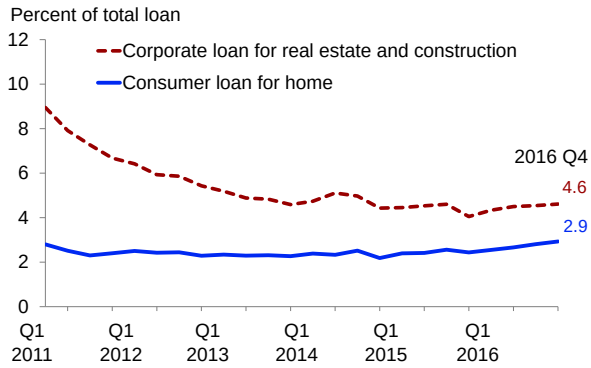
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

Real estate price indices



Source: Bank of Thailand

Non-performing loan (NPL)

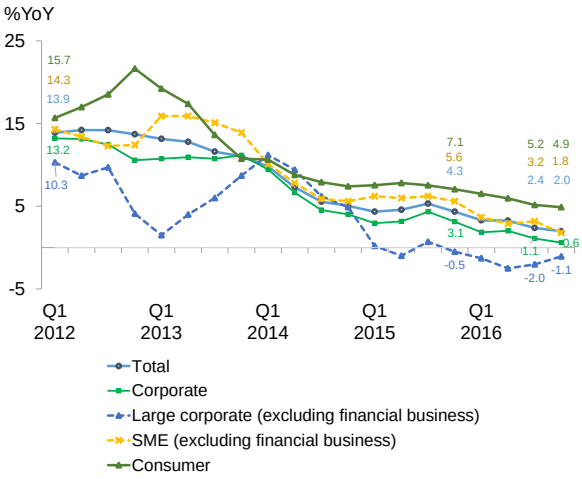


Source: Bank of Thailand

Stability: financial institutions

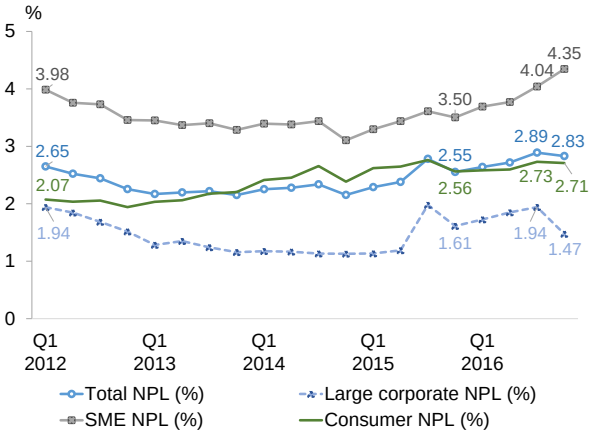
Credit growth slowed down while loan quality continued to deteriorate. However, the financial system remained sound with high levels of provisions and capital buffers.

Credit growth in the commercial bank system



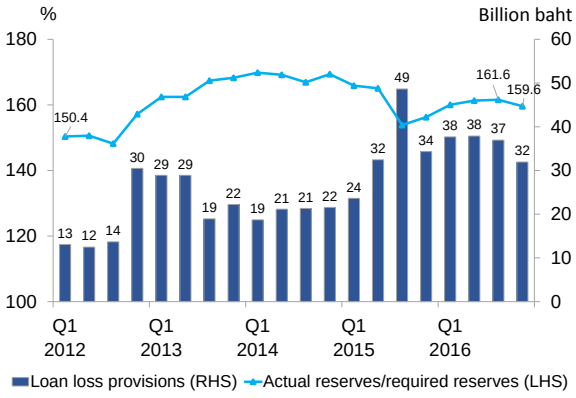
Source: Bank of Thailand

Non-performing loan (NPL)



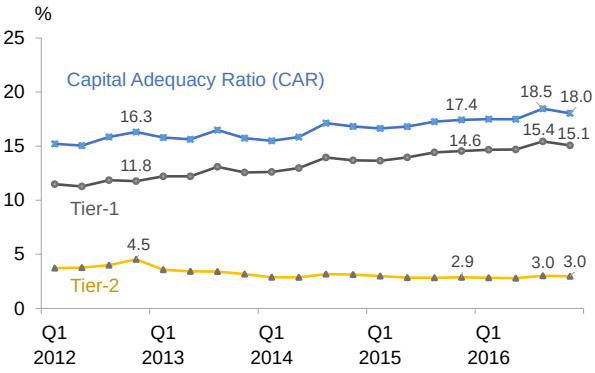
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

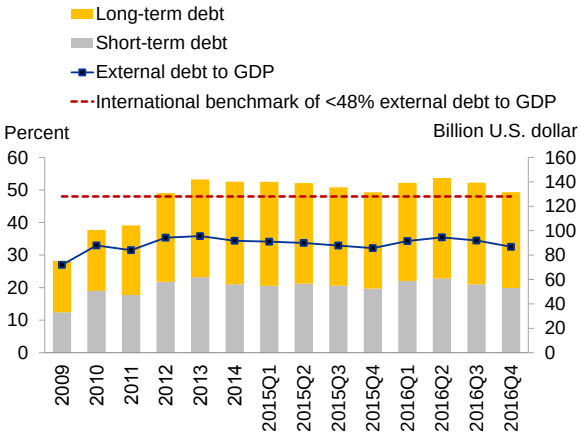


Source: Bank of Thailand

Stability: External position

Thailand's external stability remained strong with ample liquidity and high debt repayment ability.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

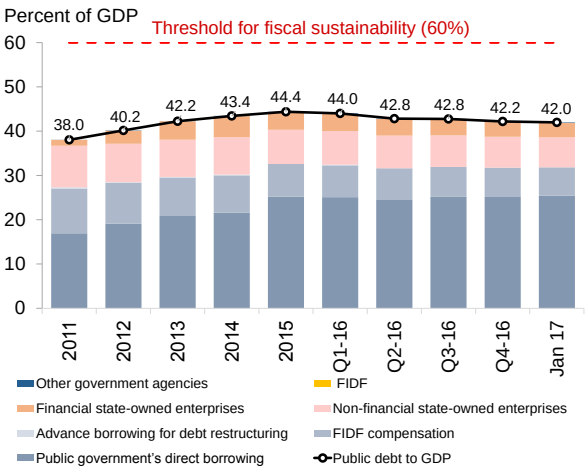


Source: Bank of Thailand

Stability: fiscal sector

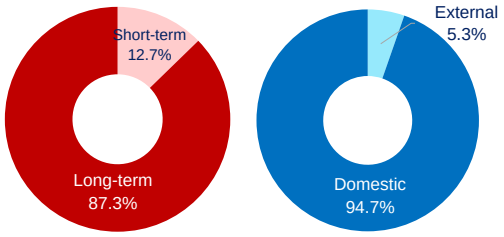
Fiscal stability remained sound, as reflected by the ratio of public debt to GDP that stayed below the fiscal sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of January 2017



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office

Pursuing Sustainable
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